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Refrigerator

Abstract

A refrigerator includes a cabinet, a door opening/closing the cabinet and having an opening that is penetrated in a front/rear direction, a transparent panel assembly mounted to cover the opening and through which the inside of the refrigerator is seen, and a lighting member provided in the door or the cabinet to brighten a rear side of the transparent panel assembly. The transparent panel assembly includes a front panel defining an outer appearance of a front surface, a rear panel defining an outer appearance of a rear surface, a spacer made of a metal material and disposed between the front panel and the rear panel to define a periphery of the transparent panel assembly. A heater mounting part on which a heater is mounted is disposed on the spacer, and when the heater generates heat, one side of the front panel, which comes into contact with the spacer, is heated.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 16/985,964, filed on Aug. 5, 2020, which is a divisional of U.S. application Ser. No. 15/934,390, filed on Mar. 23, 2018, now U.S. Pat. No. 10,767,917, which claims the benefit of an earlier filing date and right of priority under 35 U.S.C. 119 and 35 U.S.C. 365 to Korean Patent Application No. 10-2017-0037839, filed on Mar. 24, 2017 and Korean Patent Application No. 10-2017-0166450, filed on Dec. 6, 2017, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

- (1) The present disclosure relates to a refrigerator.
- (2) In general, a refrigerator refers to a home appliance in which food may be stored in an internal storage space, which is shielded by a door, at a low temperature. To achieve this, the refrigerator is configured to accommodate the stored food in an optimum state by cooling the internal storage space using cold air generated through heat exchange with a refrigerant circulating in a refrigeration cycle.
- (3) In recent years, refrigerators have become increasingly multi-functional with changes of dietary lives and gentrification of products, and refrigerators having various structures and convenience devices for convenience of users and for efficient use of internal spaces have been released.
- (4) The storage space of the refrigerator may be opened/closed by the door. Further, the refrigerator may be classified into various types according to arrangement of the storage space and a structure of the door configured to open/close the storage space.
- (5) In general, the refrigerator has a problem in that when the door is not opened, internal food cannot be identified. That is, the door should be opened to identify whether desired food is received in a space in the refrigerator or in a separate storage space provided in the door. Further, when a user does not exactly know where the food is stored, an opening time of the door may increase or the number of times the door is opened may increase. At this time, unnecessary outflow of cold air may occur.
- (6) In recent years, to solve the above-described problem, a refrigerator in which a portion of a door is transparent or an interior of the refrigerator may be viewed, has been developed.

SUMMARY

- (7) Embodiments provide a refrigerator which is capable of preventing dew condensation from being generated on a transparent panel assembly through which the inside of the refrigerator is seen.
- (8) Embodiments also provide a refrigerator which is capable of effectively preventing dew condensation from being generated on a transparent panel assembly provided on a door so as to see the inside of the door.
- (9) Embodiments also provide a refrigerator which is capable of preventing dew condensation from being generated on a surface by heating a non-insulation region of a refrigerator door provided with a transparent panel assembly.
- (10) Embodiments also provide a refrigerator in which a heater is easily disposed to improve assembly workability and productivity.
- (11) Embodiments also provide a refrigerator in which a heater is improved in heat transfer efficiency to reduce power consumption.
- (12) Embodiments also provide a refrigerator in which a heater is mounted on an outer spacer for maintaining a distance between panels to directly heat a dew condensation generation area.
- (13) In one embodiment, a refrigerator includes: a cabinet; a door opening/closing the cabinet and having an opening that is penetrated in a front/rear direction; a transparent panel assembly which is mounted to cover the opening and through which the inside of the refrigerator is seen; and a lighting member provided in the door or the cabinet to brighten a rear side of the transparent panel assembly, wherein the transparent panel assembly includes: a front panel defining an outer appearance of a front surface; a rear panel defining an outer appearance of a rear surface; a spacer made of a metal material and disposed between the front panel and the rear panel to define a periphery of the transparent panel assembly, wherein a heater mounting part on which a heater is mounted is disposed on the spacer, and when the heater generates heat, one side of the front panel, which comes into contact with the spacer, is heated.
- (14) The spacer may be made of an aluminum alloy material.
- (15) The heater mounting part may protrude to the outside of the spacer to continuously extend in a longitudinal direction of the spacer.
- (16) A heater groove into which the heater is inserted may be defined in a protruding end of the heater mounting part.
- (17) The heater mounting part may be disposed between the front panel and the protruding end of the rear panel, and a sealant may be filled into a space between the front panel, the rear panel, and the heater mounting part to a height corresponding to that of the heater mounting part.
- (18) The heater mounting part may be exposed through a peripheral surface of the transparent panel assembly.
- (19) The heater mounting part may be disposed on an end of the spacer, which comes into contact with the front plate, and a heater groove into the heater is inserted may be defined in the heater mounting part.
- (20) An insulation space that is in a sealed state may be provided between the front panel and the rear panel, an insulator may be disposed in a periphery of the door outside the transparent panel assembly, and the spacer may be disposed in a non-insulation region between the insulation space and the insulator.
- (21) The spacer may include an outer spacer coming into contact with the front panel and the rear panel to support the front panel and the rear panel.
- (22) The door may include: an outer plate defining a front surface of the door and having a plate opening that is covered by the front panel; a door liner defining a rear surface of the door and having a liner opening that is covered by the rear panel; and a support frame disposed along the plate opening and extending toward the transparent panel assembly so that a protruding end of the front panel is seated.
- (23) A spacer protrusion protruding outward and coupled to the support frame by a coupling

member may be disposed on the spacer.

(24) The spacer protrusion and the heater mounting part may be integrated with each other.

(25) The spacer protrusion and the heater mounting part may extend along the spacer in a state of being spaced apart from each other.

(26) The spacer protrusion and the heater mounting part may protrude at the same height, and a sealant may be applied at the same height as each of the spacer protrusion and the heater mounting part between the front panel and the rear panel and between the spacer protrusion and the heater mounting part.

(27) A coupling hole to which the coupling member is coupled may be defined in the spacer protrusion, a heater groove into which the heater is inserted may be defined in the heater mounting part, and the coupling hole and the heater groove may be exposed between the sealants.

(28) The heater mounting part may be disposed further rearward than the spacer protrusion and the support frame, and in the state in which the spacer protrusion and the support frame are coupled to each other, the heater mounting part may be exposed to the outside.

(29) A display for outputting a screen may be disposed on a rear surface of the front panel, and a light guide plate supported by the spacer may be disposed at a rear side of the display.

(30) An intermediate panel may be further disposed between the front panel and the rear panel, and an additional spacer may be further provided between the front panel and the intermediate panel and between the rear panel and the intermediate panel inside the spacer to support the front and intermediate panels and the rear and intermediate panels.

(31) The intermediate panel may be provided in plurality, which are spaced apart from each other, and an additional spacer may be further provided between the plurality of intermediate panels to support the plurality of intermediate panels.

(32) An intermediate panel may be further provided between the front panel and the rear panel, and the spacer may include: a spacer which is disposed between the front panel and the intermediate panel to support the front panel and the intermediate panel and on which the heater mounting part is disposed, and a spacer disposed between the rear panel and the intermediate panel to support the rear panel and the intermediate panel.

(33) The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

(2) FIG. 1 is a front view illustrating a refrigerator according to a first embodiment of the present disclosure;

(3) FIG. 2 is a perspective view illustrating the refrigerator;

(4) FIG. 3 is a perspective view illustrating a state in which a sub-door of the refrigerator is opened;

(5) FIG. 4 is a perspective view illustrating a state in which a main door of the refrigerator is opened;

(6) FIG. 5 is a perspective view illustrating the sub-door when viewed from the front side;

(7) FIG. 6 is a perspective view illustrating the sub-door when viewed from the rear side;

(8) FIG. 7 is an exploded perspective view illustrating the sub-door;

(9) FIG. 8 is a perspective view illustrating a transparent panel assembly according to the first embodiment of the present disclosure;

(10) FIG. 9 is an exploded perspective view illustrating the transparent panel assembly;

(11) FIG. 10 is a sectional view illustrating the transparent panel assembly;

(12) FIG. 11 is a partial perspective view illustrating an arrangement state of a display cable of the transparent panel assembly;

(13) FIG. 12 is a sectional view illustrating a state in which a sealant is applied to opposite ends of the transparent panel assembly;

(14) FIG. 13 is a sectional view illustrating a state in which a sealant is applied to upper and lower ends of the transparent panel assembly;

(15) FIG. 14 is a view illustrating a process of applying a sealant to the transparent panel assembly;

(16) FIG. 15 is a perspective view illustrating a support frame according to the first embodiment of the present disclosure when viewed from the front side;

(17) FIG. 16 is a perspective view illustrating the support frame when viewed from the rear side;

(18) FIG. 17 is a view illustrating a coupling state of part A of FIG. 16;

(19) FIG. 18 is a sectional view taken along line 18-18' of FIG. 17;

(20) FIG. 19 is a partial perspective view illustrating a side frame constituting the support frame;

(21) FIG. 20 is a partial perspective view illustrating a lower frame constituting the support frame;

(22) FIG. 21 is a cutaway perspective view illustrating a state in which an outer plate and the support frame are coupled to each other according to the first embodiment of the present disclosure;

(23) FIG. 22 is an exploded cutaway perspective view illustrating a coupling structure of the outer plate and the support frame;

(24) FIG. 23 is a cutaway perspective view taken along line 23-23' of FIG. 5;

(25) FIG. 24 is a sectional view taken along line 24-24' of FIG. 5;

(26) FIG. 25 is a cross sectional view illustrating the main door and the sub-door;

(27) FIG. 26 is a longitudinal sectional view illustrating the main door and the sub-door;

(28) FIG. 27 is an enlarged view illustrating part B of FIG. 26;

(29) FIG. 28 illustrates a state in which an interior of the refrigerator is visible through the transparent panel assembly;

(30) FIG. 29 illustrates a state in which a screen is output through the transparent panel assembly;

(31) FIG. 30 is a sectional view illustrating a door according to a second embodiment of the present disclosure;

(32) FIG. 31 is a sectional view illustrating a door according to a third embodiment of the present disclosure;

(33) FIG. 32 is a sectional view illustrating a door according to a fourth embodiment of the present disclosure;

(34) FIG. 33 is a sectional view illustrating a door according to a fifth embodiment of the present disclosure;

(35) FIG. 34 is a sectional view illustrating a door according to a sixth embodiment of the present disclosure;

(36) FIG. 35 is a sectional view illustrating a door according to a seventh embodiment of the present disclosure;

(37) FIG. 36 is a sectional view illustrating a door according to an eighth embodiment of the present disclosure;

(38) FIG. 37 is a sectional view illustrating a door according to a ninth embodiment of the present disclosure;

(39) FIG. 38 is a perspective view of the sub-door when viewed from a front side;

(40) FIG. 39 is an exploded perspective view of the sub-door;

(41) FIG. 40 is a perspective view of the transparent panel assembly according to a tenth embodiment of the present disclosure;

(42) FIG. 41 is an exploded perspective view of the transparent panel assembly;

(43) FIG. 42 is a cross-sectional view of the transparent panel assembly;

- (44) FIG. **43** is a partial perspective view illustrating an arranged state of the display cable of the transparent panel assembly;
- (45) FIG. **44** is a cross-sectional view illustrating a state in which a sealant is applied to both ends of the transparent panel assembly;
- (46) FIG. **45** is a cross-sectional view illustrating a state in which the sealant is applied to upper and lower ends of the transparent panel assembly;
- (47) FIG. **46** is a view illustrating a process of applying the sealant to the transparent panel assembly;
- (48) FIG. **47** is a cutaway perspective view illustrating a state in which an outer plate and a support frame are coupled to each other according to the tenth embodiment of the present disclosure;
- (49) FIG. **48** is an exploded cutaway perspective view illustrating a coupled structure between the outer plate and the support frame;
- (50) FIG. **49** is a cutaway perspective view taken along line **49-49'** of FIG. **38**;
- (51) FIG. **50** is a cross-sectional view taken along line **50-50'** of FIG. **38**;
- (52) FIG. **51** is a transverse cross-sectional view of the main door and the sub-door;
- (53) FIG. **52** is an enlarged view illustrating a portion C of FIG. **51**;
- (54) FIG. **53** is an enlarged view illustrating a portion D of FIG. **51**;
- (55) FIG. **54** is a longitudinal cross-sectional view of the main door and the sub-door;
- (56) FIG. **55** is an enlarged view illustrating a portion E of FIG. **54**;
- (57) FIG. **56** is an enlarged view illustrating a portion F of FIG. **54**;
- (58) FIG. **57** is a cross-sectional view of a door according to an eleventh embodiment of the present disclosure;
- (59) FIG. **58** is a cross-sectional view of a door according to a twelfth embodiment of the present disclosure;
- (60) FIG. **59** is a cross-sectional view of a door according to a thirteenth embodiment of the present disclosure;
- (61) FIG. **60** is a cross-sectional view of a door according to a fourteenth embodiment of the present disclosure;
- (62) FIG. **61** is a cross-sectional view of a door according to a fifteenth embodiment of the present disclosure;
- (63) FIG. **62** is a cross-sectional view of a door according to a sixteenth embodiment of the present disclosure;
- (64) FIG. **63** is a perspective view illustrating a refrigerator according to a seventeenth embodiment of the present disclosure;
- (65) FIG. **64** is a perspective view illustrating a refrigerator according to an eighteenth embodiment of the present disclosure; and
- (66) FIG. **65** is a perspective view illustrating a refrigerator according to a nineteenth embodiment of the present disclosure.

DETAILED DESCRIPTION

- (67) Hereinafter, detailed embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. However, the scope of the present disclosure is not limited to proposed embodiments, and other regressive inventions or other embodiments included in the scope of the spirits of the present disclosure may be easily proposed through addition, change, deletion, and the like of other elements.
- (68) FIG. **1** is a front view illustrating a refrigerator according to a first embodiment of the present disclosure. Further, FIG. **2** is a perspective view illustrating the refrigerator.
- (69) As illustrated in the drawings, an outer appearance of a refrigerator **1** according to the first embodiment of the present disclosure may be formed by a cabinet **10** defining a storage space and doors configured to open/close the storage space.
- (70) An interior of the cabinet **10** may be vertically partitioned by a barrier **11** (in FIG. **4**), a

refrigerating chamber **12** may be formed above the cabinet **10**, and a freezing chamber **13** may be formed below the cabinet **10**.

(71) Further, a control unit **14** configured to control an overall operation of the refrigerator **1** is formed on an upper surface of the cabinet **10**. The control unit **14** may be configured to control electrical components for selectively seeing through a see-through part **21** and outputting a screen as well as a cooling operation of the refrigerator **1**.

(72) The doors may include refrigerating chamber doors **20** and freezing chamber doors **30**. The refrigerating chamber doors **20** may be configured to open/close an opened front surface of the refrigerating chamber **12** through pivoting, and the freezing chamber doors **30** may be configured to open/close an opened front surface of the freezing chamber **13** through pivoting.

(73) Further, the pair of refrigerating chamber doors **20** are provided on left and right sides, and the refrigerating chamber **12** may be shielded by the pair of doors. Further, the pair of freezing chamber doors **30** are provided on left and right sides, and the freezing chamber **13** may be opened/closed by the pair of doors. Of course, the freezing chamber doors **30** may be configured to be drawable in a drawer form if necessary, and one or more freezing chamber doors **30** may be configured.

(74) Meanwhile, although an example where a French-type door that includes a pair of doors and opens/closes one space by rotating the doors is applied to a bottom freeze-type refrigerator in which a freezing chamber **13** is provided below is illustratively described in the embodiment of the present disclosure, the present disclosure may be applied to all types of refrigerators having doors regardless of types of the refrigerators.

(75) Further, depressed handle grooves **201** and **301** may be formed at a lower end of the refrigerating chamber doors **20** and an upper end of the freezing chamber doors **30**. A user inserts a hand into the handle grooves **201** and **301** to open/close the refrigerating chamber doors **20** or the freezing chamber doors **30**.

(76) Meanwhile, at least one door may be formed to see through an interior of the refrigerator **1**. The see-through part **21**, through which a storage space on a rear surface of the door and/or an internal space of the refrigerator **1** may be seen, may be formed in the refrigerating chamber door **20**. The see-through part **21** may form at least a part of the front surface of the refrigerating chamber door **20**. The see-through part **21** may be selectively transparent or opaque depending on manipulation by the user, and the user may accurately identify food accommodated in the refrigerator **1** through the see-through part **21**.

(77) Further, in the embodiment of the present disclosure, a case where the see-through part **21** is formed in the refrigerating chamber door **20** is described as an example. However, the see-through part **21** may be provided in various other types of refrigerator doors including the freezing chamber doors **30** according to the structure and shape of the refrigerator **1**.

(78) FIG. 3 is a perspective view illustrating a state in which a sub-door of the refrigerator is opened. Further, FIG. 4 is a perspective view illustrating a state in which a main door of the refrigerator is opened.

(79) As illustrated in the drawings, the right refrigerating chamber door **20** (when viewed in FIG. 3) of the pair of the refrigerating chamber doors **20** may be dually opened/closed. In detail, the right refrigerating chamber door **20** may include a main door **40** configured to open/close the refrigerating chamber **12** and a sub-door **50** rotatably arranged in the main door **40** to open/close an opening **41** of the main door **40**.

(80) The main door **40** may have the same size as that of the left refrigerating chamber door **20** (when viewed in FIG. 1) of the pair of refrigerating chamber doors **20** and may be rotatably mounted on the cabinet **10** by an upper hinge **401** and a lower hinge **402** to open/close at least a portion of the refrigerating chamber **12**.

(81) Further, an opening **41** opened to have a predetermined size is formed in the main door **40**. Door baskets **431** may be mounted on a rear surface of the main door **40** as well as inside the

opening **41**. At this time, the opening **41** may be formed to occupy most of the front surface of the main door **40** except for a portion of the periphery of the main door **40**.

(82) Further, a main gasket **45** is provided at a periphery of the rear surface of the main door **40** to prevent cold air inside the cabinet **10** from being leaked when the main door **40** is opened/closed.

(83) The sub-door **50** may be pivotably mounted on the front surface of the main door **40** to open/close the opening **41**. Thus, the opening **41** may be exposed through opening the sub-door **50**.

(84) The size of the sub-door **50** is equal to the size of the main door **40** so that the sub-door **50** may shield the entire front surface of the main door **40**. Further, in a state in which the sub-door **50** is closed, the main door **40** and the sub-door **50** are coupled to each other, so that the size and the shape of the coupled main door **40** and the sub-door is equal to the size and the shape of the left refrigerating chamber door **20**. Further, a sub-gasket **503** is provided on the rear surface of the sub-door **50** to seal a space between the main door **40** and the sub-door.

(85) A transparent panel assembly **60**, through which an interior of the refrigerator may be selectively seen and which may output a screen, is provided at a center of the sub-door **50**. Thus, even in a state in which the sub-door **50** is closed, the inner side of the opening **41** may be seen and may be output. The see-through part **21** may be defined as a portion on the sub-door **50**, through which the interior of the refrigerator **1** is seen, and may not necessarily coincide with the entire transparent panel assembly **60**.

(86) The transparent panel assembly **60** may be changed to a transparent state or an opaque state depending on manipulation by the user. Thus, only when the user wants to make the transparent panel assembly **60** be transparent, the transparent panel assembly **60** becomes transparent so that the interior of the refrigerator **1** is visualized, and when the user does not want to make the transparent panel assembly **60** be transparent, the transparent panel assembly **60** may be maintained in an opaque state. Further, the screen may be output in a state in which the transparent panel assembly **60** is in a transparent state or an opaque state.

(87) In the embodiment of the present disclosure, the transparent panel assembly **60** is configured to shield an opened portion of the sub-door **50**. However, according to types of the door, even when one door is configured as in the right door **20** of the refrigerating chamber **12**, an opening may be formed in the door **20**, and the transparent panel assembly may be mounted to shield the opening of the door **20**. That is, it is noted that the transparent panel assembly **60** may be applied to all types of doors, through which an opening is formed, regardless of the shape of the refrigerator and the shape of the door.

(88) As a sub-upper hinge **501** and a sub-lower hinge **502** are provided at an upper end and a lower end of the sub-door **50**, respectively, the sub-door **50** may be pivotably mounted on the front surface of the main door **40**. Further, an opening device **59** may be provided in the sub-door **50**, and a locking unit **42** may be provided in the main door **40** corresponding to the opening device **59**. Thus, the sub-door **50** may be maintained in a closed state by coupling between the opening device **59** and the locking unit **42**, and when the opening device **59** and the locking unit **42** is uncoupled from each other by manipulation of the opening device **59**, the sub-door **50** may be opened with respect to the main door **40**.

(89) Further, a damping device **504** (in FIG. 6) may be provided at a lower end of the sub-door **50**. The damping device **504** may be located at a lower lateral edge of the sub-door **50**, which is adjacent to the sub-lower hinge **502** such that an impact when the heavy sub-door **50** having is closed is absorbed by the transparent panel assembly **60**.

(90) Meanwhile, a storage case **43** may be provided on the rear surface of the main door **40**. The plurality of door baskets **431** may be arranged in the storage case **43**, and case doors **432** may be provided in the storage case **43**.

(91) FIG. 5 is a perspective view illustrating the sub-door when viewed from the front side. FIG. 6 is a perspective view illustrating the sub-door when viewed from the rear side. Further, FIG. 7 is an exploded perspective view illustrating the sub-door.

(92) As illustrated in the drawings, the sub-door **50** may include an outer plate **51** defining an outer appearance, a door liner **56** spaced apart from the outer plate **51**, the transparent panel assembly **60** mounted on an opening of the outer plate **51** and the door liner **56**, and an upper cap decoration **54** and a lower cap decoration **55** defining an upper surface and a lower surface of the sub-door **50**, and an outer appearance of the sub-door **50** may be defined by a combination of them.

(93) The outer plate **51**, which defines the front surface and a portion of a peripheral surface of the sub-door **50**, may be formed of a plate-shaped stainless material. The outer plate **51** may define a portion of the outer appearance of the sub-door **50** as well as the front surface of the sub-door **50**. Further, the outer plate **51** may be formed of the same material as that of the front surfaces of the refrigerating chamber door **20** and the freezing chamber door **30**. The front surface of the outer plate **51** may be subjected to various surface treatments such as anti-fingerprint coating, a hair line, coating for realizing a color or a pattern, and attachment of a film.

(94) The outer plate **51** may include a front surface part **512** defining an outer appearance of the front surface thereof, and side surface parts **513** defining outer appearances of side surfaces thereof exposed to the outside. Further, a plate opening **511** may be formed at the center of the front surface part **512**, and the plate opening **511** may be shielded by the transparent panel assembly **60**. Further, because the interior of the refrigerator **1** may be seen through the transparent panel assembly **60** configured to shield the plate opening **511**, the inner side of the plate opening **511** may be referred to as the see-through part **21**.

(95) The front surface part **512** may be formed to have a curvature such that the front surface part **512** is lowered as it goes from the central side to the outer side of the refrigerator **1**. The front surface part **512** may be rounded to correspond to the front surface of the neighboring refrigerating chamber door **20**, and an outer appearance of the front surface of the refrigerator may be overall seen to be in three dimensions.

(96) Further, a bent plate part **514** which is bent rearwards may be formed along a peripheral surface of the plate opening **511**. The bent plate part **514** may be formed along the periphery of the plate opening **511**, and may extend in a predetermined length such that the bent plate part **514** may be inserted into and fixed to a support frame **70**, which will be described below in detail. Thus, the plate opening **511** may be also defined by the bent plate part **514**.

(97) The side surface parts **513** which are bent rearwards may be formed at opposite ends of the front surface part **512**. The side surface parts **513** may define outer appearances of the side surfaces of the sub-door **50**. Further, ends of the side surface parts **513** may be bent inwards to be coupled to the door liner **56**. Further, an upper end and a lower end of the front surface part **512** may be also bent inwards to be coupled to the upper cap decoration **54** and the lower cap decoration **55**.

(98) Meanwhile, an upper end and a lower end of the outer plate **51** may be also bent, and may be coupled to the upper cap decoration **54** and the lower cap decoration **55**. Thus, the outer plate **51** may be coupled to the door liner **56**, the upper cap decoration **54**, and the lower cap decoration **55**, to define an outer appearance of the sub-door **50**.

(99) The door liner **56** defines the rear surface of the sub-door **50**, and a liner opening **561** is formed in an area in which the transparent panel assembly **60** is arranged. Further, a sub-gasket **503** configured to seal a gap between the sub-door **50** and the main door **40** may be mounted on the rear surface of the door liner **56**.

(100) Further, door lights **57** may be provided on opposite sides of the liner opening **561**. The door lights **57** may be configured to illuminate the rear surface of the sub-door **50** and the rear side of the transparent panel assembly **60**. The door lights **57** may be referred to as lighting members, and the lighting members may include another light provided inside the storage space to illuminate the interior of the refrigerator **1** as well as the door lights **57**.

(101) Thus, the door lights **57** may illuminate the internal space of the storage case **43**, and at the same time, may function as auxiliary backlights for the transparent panel assembly **60** to make the screen clearer when the screen of the transparent panel assembly **60** is output. When the door lights

57 are lighted, an interior of the storage case **43** becomes brighter. Thus, the interior of the refrigerator **1** is brighter than an exterior of the refrigerator **1**, so that a rear space of the sub-door **50** may be visualized through the transparent panel assembly **60**.

(102) The door lights **57** may be arranged on opposite sides of the transparent panel assembly **60** to face each other. The door lights **57** may be arranged at various positions as long as the rear side of the sub-door **50** may have a sufficient brightness.

(103) Further, the opening device **59** may be mounted on the door liner **56**. The opening device **59** may include a manipulation member **591** exposed to the lower end of the sub-door **50**, a rod **592** extending from the manipulation member **591**, and a locking member **593** protruding from the rear surface of the door liner **56**. The rod **592** moves the locking member **593** by manipulation of the manipulation member **592** by the user, so that the sub-door **50** is selectively restrained to the main door **40**, and opening/closing of the sub-door **50** may be manipulated.

(104) The upper cap decoration **54**, which defines an upper surface of the sub-door **50**, is coupled to upper ends of the outer plate **51** and the door liner **56**. The upper surface of the upper cap decoration **54** is opened so that a decoration opening **542** communicating with an upper space of the transparent panel assembly **60** is formed, and is shielded by a decoration cover **543**. Further, a printed circuit board (PCB) mounting part **543a** is formed in the decoration cover **543**, so that PCBs **602**, **603**, and **604** for operating electrical components inside the transparent panel assembly **60** and the sub-door **50** may be mounted on the PCB mounting part **543a**. The PCBs **602**, **603**, and **604** may be configured in at least one module form, and may be provided in a closed space on an upper side of the sub-door **50**.

(105) At this time, the space on the upper side of the sub-door **50** may be partitioned into front and rear spaces by an upper portion of the support frame **70**, an insulator **531a** may be arranged in the front space, and the PCBs **602**, **603**, and **604** may be arranged in the rear space. The structure of the space on the upper side of the sub-door **50** will be described with reference to FIG. 27.

(106) The lower cap decoration **55**, which defines a lower surface of the sub-door **50**, is coupled to lower ends of the outer plate **51** and the door liner **56**.

(107) The transparent panel assembly **60** may be arranged between the outer plate **51** and the door liner **56**. Further, the transparent panel assembly **60** may be configured to shield the plate opening **511** and tee door liner opening **561**. Further, the transparent panel assembly **60** may be selectively manipulated by the user in one of a transparent state, a translucent state, an opaque state, and a screen outputting state.

(108) Thus, the user may selectively see through the internal space of the sub-door **50** through the transparent panel assembly **60**, and may view the screen output through the transparent panel assembly **60** as well.

(109) Of course, the transparent panel assembly **60** may not include a display **62** for outputting a screen, and the transparent panel assembly **60** without the display **62** may have the same outer appearance as that of the transparent panel assembly **60** having the display **62** only with a difference in that the screen is not output.

(110) The support frame **70** configured to support the transparent panel assembly **60** is mounted on a periphery of the plate opening **511** of the outer plate **51**. The transparent panel assembly **60** may be fixed and mounted to the outer plate **51** by the support frame **70**. In particular, the front surface of the outer plate **51** and the front surface of the transparent panel assembly **60** are arranged on the same extension line, so that the front surface of the sub-door **50** may have a sense of unity.

(111) The support frame **70** has a frame opening **701** formed at a center thereof, and the frame opening **701** is formed to be slightly smaller than the plate opening **511**, so as to provide a structure on which the transparent panel assembly **60** may be seated. Further, the frame opening **701** may be formed to be smaller than a front panel **61** and to be larger than a rear panel **65**. Thus, when the transparent panel assembly **60** is mounted, the rear panel **65** may sequentially pass through the plate opening **511** and the frame opening **701**, and then may be seated on the door liner **56**.

(112) Further, the support frame **70** has a coupling structure with the outer plate **51**, and the outer plate **51** and an end of the transparent panel assembly **60** may be mounted in close contact with each other. Thus, when the sub-door **50** is viewed from the front side, an end of the outer plate **51** and a periphery of the transparent panel assembly **60** are in close contact with each other, so that a gap between the outer plate **51** and the transparent panel assembly **60** is rarely viewed or is viewed in a form of a line, and the outer appearance of the front surface may be viewed as having senses of continuity and unity.

(113) The support frame **70** supports the outer plate **51** and the transparent panel assembly **60** and, at the same time, also has a fixing structure for a heater **532** arranged on the rear surface of the transparent panel assembly **60**. Thus, the heater **532** may be arranged on the rear surface of the transparent panel assembly **60** while being mounted on the support frame **70**, and at this time, may be arranged on a bezel **611** formed along a periphery of the front panel **61**, so that structures of the heater **532** and the support frame **70** may not be exposed to the outside.

(114) Hereinafter, the structures of the transparent panel assembly and the support frame will be described in more detail.

(115) FIG. **8** is a perspective view illustrating a transparent panel assembly according to the first embodiment of the present disclosure. Further, FIG. **9** is an exploded perspective view illustrating the transparent panel assembly. Further, FIG. **10** is a sectional view illustrating the transparent panel assembly.

(116) As illustrated in the drawings, the transparent panel assembly **60** may be formed to have a size in which the transparent panel assembly **60** may shield the plate opening **511** and the liner opening **561** from the inner side of the sub-door **50**. Further, the see-through part **21** may be formed such that a space in the refrigerator **1** may be selectively visualized and the screen may be output.

(117) The transparent panel assembly **60** may be configured by a plurality of panels having a shape of a plate, and may be configured such that the panels are spaced apart from each other by at least one spacer at a specific interval. The transparent panel assembly **60** may include the front panel **61** and the rear panel **65** defining at least the front surface and the rear surface thereof, and a spacer connecting the front panel **61** and the rear panel **65** between the front panel **61** and the rear panel **65**, and may have a structure in which an additional panel and an additional spacer are further provided in an internal space defined by the spacer.

(118) The transparent panel assembly **60** will be described with reference to the drawings. The outer shape of the transparent panel assembly **60** may be defined by the front panel **61** and the rear panel **65** defining the front surface and the rear surface of the transparent panel assembly **60**, and an outer spacer **67** connecting the front panel **61** and the rear panel **65** to each other.

(119) Further, between the front panel **61** and the rear panel **65**, the display **62** and a light guide plate **64** may be arranged, a first spacer **63** configured to support the display **62** and the light guide plate **64** may be further provided, and display lights **68** configured to irradiate light to the light guide plate **64** may be provided.

(120) In more detail, the front panel **61**, which defines an outer appearance of the front surface of the transparent panel assembly **60**, may be formed of transparent glass (for example, blue glass). Of course, the front panel **61** may be formed of another material through which the interior of the refrigerator may be seen and a touch input may be performed.

(121) Further, a film, through which light selectively passes depending on an ON/OFF state of a light inside the refrigerator **1** or a light provided in the sub-door **50** so that the film may be selectively transparent or opaque, may be arranged on the rear surface of the front panel **61**.

(122) The front panel may be formed to have a size corresponding to the size of the plate opening **511**, and may be formed to be larger than the size of the frame opening **701**. Thus, the periphery of the front panel **61** may be supported by the support frame **70**. Further, in a state in which the transparent panel assembly **60** is mounted, an end of the front panel **61** may be in contact with an end of the plate opening **511**, and a space may not be formed between the plate opening **511** and the

front panel **61**.

(123) In detail, the front panel **61** may have a front protrusion **613** formed therein to protrude more outward than the rear panel **65**. Due to structural characteristics of the front protrusion **613** inserted into and mounted on the front side of the outer plate **51**, the front protrusion **613** may protrude more upward/downward/leftward/rightward than the rear panel **65** and the outer spacer **67**. Thus, the front panel **61** defining the front surface of the transparent panel assembly **60** may further extend outward the frame opening **701**, and thus may be stably supported by the support frame **70**. The rear panel **65** and the like as well as the outer spacer **67** may be inserted into the frame opening **701**.

(124) Further, the support frame **70** and the outer spacer **67** of the transparent panel assembly **60** may be fastened and coupled to each other through a separate coupling structure or coupling members **78** such as a screw. Thus, when the transparent panel assembly **60** is mounted, the front protrusion **613** may be supported by the support frame **70**, and at the same time, the support frame **70** may be coupled to the outer spacer **67**, so that the heavy transparent panel assembly **60** may be maintained in a stably fixed and mounted state even when the sub-door **50** is opened/closed.

(125) Meanwhile, the bezel **611** may be formed along a periphery of the rear surface of the front panel **61**. The bezel **611** may be formed by printing with an opaque color such as black, and may be formed to have a predetermined width such that the outer spacer **67**, the first spacer **63**, the heater **532**, and the like may be covered without being exposed to the outside. The bezel **611** may be formed to have a width from an outer end of the front panel **61** to the first spacer **63**.

(126) A touch sensor **612** may be arranged on the rear surface of the front panel **61**. The touch sensor **612** may be formed on the rear surface of the front panel **61** in a printing scheme, and may be configured to detect a touch operation on the front panel by the user. Of course, the touch sensor **612** may employ various other schemes such as a film bonding scheme not the printing scheme, in which input may be performed through a touch on the front panel **61**.

(127) A touch cable **601** connected to the touch sensor **612** may be provided at an upper end of the front panel **61**. The touch cable **601** may be a flexible film type cable such as a flexible flat cable (FFC) and a flexible print cable or a flexible print circuit board (FPC), and a printed circuit may be printed on the touch cable **601** to form at least a portion of a touch PCB **603**. Further, the touch cable **601** may be connected to the touch PCB **603** provided above the sub-door **50**.

(128) The touch cable **601** may be connected to the touch sensor **612** and may extend upwards. Further, the touch cable **601** may be configured such that wires are arranged in a base, such as a film, formed of resin, and may upwards extend along the rear surface of the front panel **61**. The touch cable **601** may be formed to have a thin thickness and a wide width, which is similar to a sheet, and thus may be flexibly bent.

(129) Further, the touch cable **601** may be configured in a film type, and may have a structure in which an end of the touch cable **601** is easily inserted into a connector of the touch PCB **603** when the touch cable **601** is connected to the touch PCB **603**. To achieve this, the touch cable **601** may be bent several times, and the end of the touch cable **601** may be formed toward the connector on the touch PCB **603**. Further, the touch cable **601** is bent to be arranged along a wall surface of the internal space of the sub-door **50**, so that the space inside the sub-door **50** may be efficiently arranged.

(130) Further, in addition to the touch cable **601**, display cables **605** and display light cables **606** may be formed to have the same structure. In this way, all the cables **601**, **605**, and **606** formed to have a flat cable shape may extend to an upper end of the transparent panel assembly **60**, and may be efficiently arranged on the sub-door **50** having a thin thickness and a wide width. In addition, the cables **601**, **605**, and **606** may provide a simple connection structure with the PCBs **602**, **603**, and **604** arranged above the sub-door **50**.

(131) Meanwhile, the display **62** may be provided on the rear surface of the front panel **61**. The display **62** may be a liquid crystal display (LCD) module configured to output a screen, and may be

transparent to be seen through in a state in which the screen is not output.

(132) Source boards **621** may be provided at one end of opposite left and right ends of the display **62**. The source boards **621**, which are adapted to output the screen of the display **62**, may be formed in an assembly state while being connected to the display **62**. Further, portions of the source boards **621** may also include a flexible film type cable structure.

(133) Further, the widths of the source boards **621** may be smaller than the thickness of the transparent panel assembly **60**, and may be bent while the transparent panel assembly **60** is assembled. At this time, the source boards **621** may be arranged between the outer spacer **67** and the first spacer **63**, and may be in contact with an inner surface of the outer spacer **67** while being perpendicular to the front panel **61**.

(134) Further, the source boards **621** may be connected to the display cables **605**, and the display cables **605** may be connected to the T-CON board **602** above the sub-door **50**.

(135) In detail, when the source boards **621** are arranged on the rear surface of the display **62**, the source boards **621** may be exposed to the outside through the see-through part **21** due to characteristics of the display **62** which is transparent. Further, when the source boards **621** have a structure protruding sideward, there is a problem in that the size of the sub-door **50** is enlarged.

(136) Thus, the source boards **621** may be formed at a peripheral end of the display **62**, and may be provided between the outer spacer **67** and the first spacer **63**. Further, the source boards **621** may be formed to have a size corresponding to the outer spacer **67** so as not to depart from the outer spacer **67** in a state in which the source boards **621** are in close contact with the outer spacer **67**.

(137) Meanwhile, the two upper and lower source boards **621** may be formed, and may be connected to the pair of display cables **605**, respectively. The display cables **605** may have a flexible and flat structure, which is similar to the touch cable **601**, and may have a freely-bent structure.

(138) The display cables **605** may extend along a peripheral surface of the transparent panel assembly **60**, and may pass through a sealant **608** defining the peripheral surface of the transparent panel assembly **60** to extend to the outside of the transparent panel assembly **60**.

(139) Further, the display cables **605** may be bent to extend along the peripheral surface of the transparent panel assembly **60**, and may be bent such that ends of the display cables **605** may extend upwards. Thus, the display cables **605** may be coupled to the T-CON board **602** above the sub-door **50**.

(140) Meanwhile, opposite ends of the display **62** may be supported by the first spacer **63**. The first spacer **63** may be formed to have a rod shape extending from an upper end to a lower end of the display **62**, and may be formed of aluminum.

(141) The light guide plate **64** may be located behind the display **62**, and may be spaced apart from the display **62** by a predetermined distance by the first spacer **63**. Here, a sense of depth of the screen output on the display **62** may differ according to the position of the light guide plate **64**.

(142) The light guide plate **64**, which is adapted to diffuse or scatter light irradiated by the display lights **68**, may be formed of various materials. For example, the light guide plate **64** may be formed of polymer, and may be formed such that a pattern is formed on a surface of the light guide plate **64** or a film is attached to the surface of the light guide plate **64**. The light guide plate **64** is configured to illuminate the display **62** on the rear side in a state in which the display lights **68** are turned on. To achieve this, the light guide plate **64** may be formed to have a plate shape having a size that is equal to or slightly larger than the size of the display **62**, and the display lights **68** may be provided at locations corresponding to an upper end and a lower end of the light guide plate **64**.

(143) Of course, when the display **62** is not provided, a separate glass or a heat insulating glass instead of the light guide plate **64** may be arranged.

(144) The rear panel **65** may be arranged behind the light guide plate **64**. The rear panel **65**, which defines the rear surface of the transparent panel assembly **60**, may be formed to be larger than the light guide plate **64** and to be smaller than the front panel **61**. Further, the rear panel **65** may be

formed to be larger than the liner opening **561**, and may shield the liner opening **561**.

(145) Meanwhile, the periphery of the rear panel **65** may protrude more outward than the outer spacer **67**, to form a rear panel protrusion **651**. The rear panel protrusion **651** may have a protruding portion which may be seated on the door liner **56** when the transparent panel assembly **60** is mounted, and may define a space in which the sealant applied to the periphery of the sub-door **50** may be filled.

(146) For insulation, the rear panel **65** may be formed of low- ϵ glass. Thus, the rear panel **65** may prevent cold air in the refrigerator **1** from being heat-exchanged with the outside through the transparent panel assembly **60**.

(147) A pair of second spacer **66** may be provided between the rear panel **65** and the light guide plate **64**. The second spacers **66** may be formed to have a shape of a quadrangular frame formed along the periphery of the light guide plate **64**, and may adhere to the light guide plate **64** and the rear panel **65** so that the light guide plate **64** and the rear panel **65** may be spaced apart from each other by a predetermined distance. Further, a heat insulating glass **69** may be provided between the pair of second spacer **66**. A multilayered insulating layer may be provided between the light guide plate **64** and the rear panel **65** by the heat insulating glass **69**. Of course, a structure in which the light guide plate **64** and the rear panel **65** are fixed to each other by one second spacer **66** without the heat insulating glass **69** may be adopted as needed.

(148) In the embodiment of the present disclosure, all the spacers **63**, **66**, and **67** have different structures, but perform support to maintain an interval between the neighboring panels **61** and **65** or the light guide plate **64**. Further, various forms such as a rod and a form in which a moisture absorbent is accommodated may be applied to the spacers **63**, **66**, and **67**.

(149) The interval between the front panel **61** and the light guide plate **64** is maintained at a fixed interval to output the screen of the display **62**. Further, the interval between the light guide plate **64** and the rear panel **65** may be determined based on the thickness of the sub-door **50** or the entire thickness of the transparent panel assembly **60**. That is, as the thickness of the second spacers **66** is adjusted, the entire thickness of the transparent panel assembly **60** is determined, and thus the transparent panel assembly **60** may be mounted in accordance with the specification of the sub-door **50**.

(150) Meanwhile, the rear panel **65** may be in contact with the door light **57**, and a distance between the display **62** and the door lights **57** may be determined based on the position of the rear panel **65**. A space behind the transparent panel assembly **60** may be illuminated by the door lights **57**, making it possible to visualize the storage space. Further, the door lights **57** may function as auxiliary backlights of the display **62** in a lit state.

(151) A space between the light guide plate **64** and the rear panel **65** may be sealed by the second spacers **66**. Thus, a space between the second spacers **66** and the light guide plate **64** is made to be in a vacuum state or an adiabatic gas for insulation, such as argon, is injected into the space, so that insulation performance may be further improved.

(152) In a state in which the rear panel **65** adheres to the second spacers **66**, an outer end of the rear panel **65** may extend more outward than the second spacers **66**. Further, the outer spacer **67** is mounted to the outer end of the rear panel **65**, the rear panel **65** and the front panel **61** may be fixed to each other.

(153) The outer spacer **67** may be formed to have a shape of a rectangular frame, and the outer spacer **67** may connect the rear surface of the front panel **61** and the front surface of the rear panel **65** to each other, and at the same time, may define the peripheral surface of the transparent panel assembly **60**.

(154) In detail, the outer spacer **67** defines a periphery of an outer portion of the transparent panel assembly **60**, and at the same time, has a structure for connecting the front panel **61** at a specific interval.

(155) A space between the front panel **61** and the rear panel **65**, that is, an internal space of the

outer spacer **67**, may be completely sealed by coupling of the outer spacer **67**. Further, the inside of the outer spacer **67** may be further sealed by the sealant **608** applied to the periphery of the outer spacer **67**.

(156) The display **62** and the light guide plate **64** may be spaced forward/rearward apart from each other in the space sealed by the outer spacer **67**, and the first spacer **63** and the second spacers **66** for maintaining the interval of the light guide plate **64** may be also provided in the internal space of the outer spacer **67**.

(157) Of course, an additional insulation panel or a multilayered glass structure may be further provided inside the outer spacer **67**, and these configurations may be provided inside the space defined by the outer spacer **67**.

(158) That is, the overall appearance of the transparent panel assembly **60** may be defined by the front panel **61**, the rear panel **65**, and the outer spacer **67**, and all the other configurations may be provided inside the outer spacer **67**. Thus, only the spaces between the outer spacer **67**, the front panel **61**, and the rear panel **65** are sealed, so that the multilayered panel structure may be completely sealed.

(159) In particular, even when a plate-shaped structure as well as the light guide plate **64** is further provided inside the outer spacer **67**, if only the outermost outer spacer **67** adheres to the front panel **61** and the rear panel **65**, a sealing structure of the transparent panel assembly **60** may be completed. Such a sealing structure may maintain the minimum sealing points even in the multilayered structure by a plurality of panels including the light guide plate **64**.

(160) Thus, a probability that external air is introduced into the transparent panel assembly **60** or dew is condensed inside the transparent panel assembly **60** due to moisture permeation may be minimized. Further, the inside of the outer spacer **67** is made to be in a vacuum state or a gas for insulation is injected into the outer spacer **67**, a heat insulating layer may be formed in the entire multilayered structure inside the transparent panel assembly **60**, thereby further improving insulation performance.

(161) As a result, as the transparent panel assembly **60** is arranged inside the sub-door **50**, the interior of the refrigerator may be seen, the screen may be output, and an insulation structure may be completed in the multilayered panel structure, so that insulation performance may be ensured.

(162) Further, a space on which the display lights **68** may be mounted may be provided on an inner surface of the outer spacer **67**. The display lights **68** may be mounted at an upper end and a lower end of the outer spacer **67**, and the light guide plate **64** may be located between the display lights **68** arranged at the upper end and the lower end of the outer spacer **67**.

(163) Thus, light irradiated by the display lights **68** may be directed toward an end of the light guide plate **64**, and may be moved along the light guide plate **64** so that the light guide plate **64** may emit light from the entire surface thereof.

(164) Meanwhile, the display lights **68** located at an upper end and a lower end of the inside of the transparent panel assembly **60** may be connected to the display light cables **606**. The display light cables **606** may be formed to have a flexible and flat shape, which is like the touch cable **601** and the display cables **605**.

(165) The display light cables **606** may be connected to the display lights **68** mounted inside the outer spacer **67** and may extend toward the outside of the transparent panel assembly **60**.

(166) Further, the display light cables **606** may extend along a periphery of the transparent display **62** so as not to be exposed through the transparent display **62**. Further, the display light cables **606** may extend upwards while being in close contact with the rear panel **65**, and may be bent while being in contact with the rear surface of the rear panel **65**, to be connected to the docking PCB **604** above the sub-door **50** as needed.

(167) Here, the display light cables **606** extends while being in close contact with the peripheral surface of the rear panel protrusion **651** of the rear panel **65**, and thus is not exposed through the transparent panel assembly **60** when viewed from the outside of the sub-door **50**.

(168) The sealant **608** may be applied to a periphery of the outer spacer **67**. The sealant **608** may be applied to form the peripheral surface of the transparent panel assembly **60**, and forms a peripheral surface between the front panel **61** and the rear panel **65**.

(169) The sealant **608**, which performs sealing to prevent air from being introduced into the transparent panel assembly **60**, may be formed of polysulfide (referred to as “thiokol”). Of course, if necessary, the sealant **608** may be formed of other sealant materials such as silicone and urethane which may be directly in contact with foam liquid injected to form the insulator **531**.

(170) By the sealant **608**, the coupling between the outer spacer **67**, the front panel **61**, and the rear panel **65** may be maintained, and at the same time, connection portions between components may be completely sealed, so that moisture may be prevented from being introduced. Further, the sealant **608**, which is a portion directly in contact with the foam liquid when the insulator **531** is formed, may protect the periphery of the transparent panel assembly **60**.

(171) Further, the cables **601**, **605**, and **606** connected to the touch sensor **612**, the display panel **62**, and the display lights **68** inside the transparent panel assembly **60** may be input/output through the sealant **608**. That is, the sealant **608** may block outer surfaces of the cables **601**, **605**, and **606** when the cables **601**, **605**, and **606** extend to the outside through the peripheral surface of the transparent panel assembly **60**, to prevent water or moisture from being introduced into a space through which the cables **601**, **605**, and **606** are input/output.

(172) FIG. **11** is a partial perspective view illustrating an arrangement state of a display cable of the transparent panel assembly.

(173) As illustrated in the drawing, the display cables **605** may be connected to the source boards **621** to extend upwards, may extend along a periphery of the side surface of the transparent panel assembly **60**, and then may be connected to the T-CON board **602**.

(174) The display cables **605** may be connected to the source boards **621** inside the transparent panel assembly **60**, and may be guided to the outside of the outer spacer **67** through a space between the rear panel **65** and the outer spacer **67**.

(175) In detail, cable connectors **605a** may be formed in the display cables **605**. The cable connectors **605a** may be introduced into the transparent panel assembly **60** in a space between the rear panel **65** and an end of the outer spacer **67**, and may be connected to the source boards **621** in an internal space of the transparent display **62**.

(176) The cable connectors **605a** may be guided to an outer surface of the transparent panel assembly **60** through a space between a gap of an adhesive member **671** allowing the rear panel **65** and the outer spacer **67** to adhere to each other and the sealant **608**. Thus, the display cables **605** may pass through the sealed periphery of the sealed transparent panel assembly to be guided to the outside.

(177) In this state, the display cables **605** may extend upwards in a bent state to be in contact with the outer surface of the transparent assembly **60** to which the sealant **608** is applied, and may be bent again to be connected to the T-CON board **602**. That is, the display cables **605** may extend to be connected to the T-CON board **602** while being exposed to the outside of the transparent panel assembly **60**.

(178) FIG. **12** is a sectional view illustrating a state in which a sealant is applied to opposite ends of the transparent panel assembly. Further, FIG. **13** is a sectional view illustrating a state in which a sealant is applied to upper and lower ends of the transparent panel assembly. Further, FIG. **14** is a view illustrating a process of applying a sealant to the transparent panel assembly.

(179) As illustrated in the drawings, the sealant **608** may be applied to the periphery of opposite left and right surfaces and upper and lower surfaces of the transparent panel assembly **60**. The sealant **608** may be applied to a gap between the front panel **61** and the rear panel **65**, and may be configured to cover the outer side of the outer spacer **67**.

(180) The transparent panel assembly **60** may be mounted in a state in which the sealant **608** is applied, and may be supported by the support frame **70**. Thus, there is a problem in that when the

sealant **608** does not have a uniform surface, if the transparent panel assembly **60** is assembled, the transparent panel assembly **60** may be incorrectly assembled by interference with the support frame **70** or other neighboring components or a failure may occur.

(181) In particular, when an interval between the front panel **61** and the rear panel **65** is large, it is not easy to uniformly apply the sealant **608**, and the sealant **608** may be biased to one side or may have an uneven surface in a local section.

(182) To prevent such a problem, a spacer protrusion **672** may be formed on an outer surface of the outer spacer **67**. The spacer protrusion **672** may be located at the center in the widthwise direction of the outer spacer **67**, and may extend along the lengthwise direction of the outer spacer **67**. The spacer protrusion **672** may continuously extend from one end to the other end of the outer spacer **67**, and if necessary, the spacer protrusions **672** having a specific length may be continuously arranged at a specific interval.

(183) Further, the spacer protrusion **672** may protrude to a height corresponding to the height of the rear panel **65**. Thus, the space between the front panel **61** and the rear panel **65** may be partitioned into two spaces by the spacer protrusion **672**, and the sealant **608** may be filled in the two spaces.

(184) Meanwhile, as illustrated in FIG. **14**, to allow the sealant **608** to have a uniform height, after the sealant **608** is filled in spaces **673** on opposite sides of the spacer protrusion **672**, the level of the sealant **608** may be adjusted using a separate jig or a scraper S.

(185) In detail, when the jig or the scraper S comes into contact with the peripheral surface of the transparent panel assembly **60** in a state in which the sealant **608** is filled in opposite sides of the spacer protrusion **672**, a lower end of the jig or the scraper S comes into contact with a protruding upper surface of the spacer protrusion **672** and an end of the rear panel **65**, which has the same height as that of the upper surface of the spacer protrusion **672**. Further, the other side of the jig or the scraper S is in contact with the rear surface of the front panel **61**, and in this state, when the jig or the scraper S moves, the sealant **608** is filled in the spaces on the opposite sides of the spacer protrusion **672** by the height of the spacer protrusion **672** and the rear panel **65**, and the remaining portion may be removed by the jig or the scraper S.

(186) Thus, when the jig or the scraper S moves along the periphery of the transparent panel assembly **60**, the sealant **608** may be applied to the periphery of the transparent panel assembly **60** at a uniform height. Further, when the transparent panel assembly **60** is mounted, the sealant **608** may not interfere with the support frame **70** or other components.

(187) After the sealant **608** is applied, the spacer protrusion **672** may be exposed to the peripheral surface of the transparent panel assembly **60**. Further, a plurality of coupling holes **672a** may be formed on the exposed outer surface of the spacer protrusion **672**. The plurality of coupling holes **672a**, to which the coupling members **78** are fastened for coupling with the transparent panel assembly **60**, may be formed along the spacer protrusion **672**. It is preferable that the coupling holes **672a** are arranged along the spacer protrusion **672**, and are located at a lower portion of the outer spacer **67**, which is not interfered by the cables **605**.

(188) Meanwhile, as illustrated in FIGS. **12** and **13**, the spacer protrusion **672** may be formed at the periphery on the opposite left and right surfaces and the upper and lower surface of the transparent panel assembly **60**. Thus, the sealant **608** may be applied to the entire periphery of the transparent panel assembly **60**, and upper, lower, left, and right portions of the periphery of the transparent panel assembly **60** may be stably fixed to the support frame **70**.

(189) Further, although a structure in which the spacer protrusion **672** is arranged in one row between the front panel **61** and the rear panel **65** is illustrated, if necessary, the spacer protrusion **672** may be configured in a plurality of rows.

(190) FIG. **15** is a perspective view illustrating a support frame according to the first embodiment of the present disclosure when viewed from the front side. Further, FIG. **16** is a perspective view illustrating the support frame when viewed from the rear side.

(191) As illustrated in the drawings, the support frame **70** may be injection-molded using plastic, is

formed to have a rectangular frame shape, and has a frame opening **701** formed at the center thereof. Further, the support frame **70** may be formed to have a predetermined width, and may be configured to fix the outer plate **51** and, at the same time, support the transparent panel assembly **60**.

(192) The support frame **70** may include an upper frame **71** defining an upper portion thereof, and a lower frame **72** defining a lower portion thereof, and side frames **73** connecting opposite ends of the upper frame **71** and the lower frame **72**.

(193) The entire shape of the support frame **70** having a rectangular frame shape may be formed by coupling the upper frame **71**, the lower frame **72**, and the side frames **73** to each other. In this way, the support frame **70** may be formed by coupling a plurality of components, and thus the components having relatively complex structures may be easily formed.

(194) Meanwhile, the upper frame **71** defines an upper shape of the support frame **70**, and may partition an upper space of the sub-door **50** into front and rear spaces. That is, a frame barrier **711** extending to the upper surface of the sub-door **50** may be formed in the upper frame **71**, and a space above the sub-door **50** may be partitioned into front and rear spaces by the frame barrier **711**.

(195) Further, side barriers **712** may be formed at opposite left and right ends of the frame barrier **711**. Thus, the upper side of the sub-door **50** may be partitioned into front and rear spaces by the upper frame **71**, and an independent space in which the PCBs **602**, **603**, and **604** may be accommodated may be provided in the rear space. Further, the space in which the PCBs **602**, **603**, and **604** are accommodated may communicate with the decoration opening **542** of the upper cap decoration **54**. Further, a space in which the insulator **531a** is accommodated may be formed in the front space.

(196) The lower frame **72** may be coupled to lower ends of the side frames **73**, and may be configured to support a lower portion of the outer plate **51** and the lower end of the transparent panel assembly **60**.

(197) The side frames **73** define opposite left and right sides of the support frame **70**, and vertically extends to connect the upper frame **71** and the lower frame **72** to each other between the upper frame **71** and the lower frame **72**. That is, the side frames **73** may be coupled to opposite ends of the upper frame **71** and the lower frame **72**.

(198) The entire structure may be configured to have a rectangular frame shape by such coupling between the upper frame **71**, the lower frame **72**, and the side frames **73**. Further, in a state in which the support frame **70** is assembled, the side frames **73**, the upper frame **71**, and the lower frame **72** are in contact with an end of the plate opening **511** of the outer plate **51** to support the outer plate **51**. Further, the side frames **73**, the upper frame **71**, and the lower frame **72** may be configured to support the peripheral surface of the transparent panel assembly **60**.

(199) Further, the opposite left and right ends of the upper frame **71** and the lower frame **72** may extend to the side frames **73**, and at this time, the extending portions have a shape corresponding to a sectional shape of the side frames **73**, so that a sense of unity is achieved when the frames **71**, **72**, and **73** are coupled. Thus, a coupling structure of the side frames **73**, the upper frame **71**, and the lower frame **72** may be easily formed.

(200) In the present embodiment, the support frame **70** is formed by separately forming four parts and then coupling the four parts to each other. However, if necessary, the support frame **70** may be formed by coupling two or more components.

(201) Meanwhile, the support frame **70** has a structure configured to support the outer plate **51** and the front panel **61**. In this structure, the upper frame **71**, the lower frame **72**, and the side frames **73** have the same structure.

(202) Hereinafter, a description will be made based on a structure of the side frames **73**, and the same structure may be applied to the upper frame **71** and the lower frame **72**.

(203) The support frame **70** may entirely include a plate support **74**, a plate accommodating groove **75**, a panel support **76**, and a heater accommodating groove **761**.

(204) The plate support **74**, which defines the outermost side of the support frame **70**, may have a front surface having a flat surface shape, and may be formed to be in close contact with the rear surface of the outer plate **51**. That is, the outermost periphery of the support frame **70** may support the rear surface of the outer plate **51**, and may adhere to the rear surface of the outer plate **51** through an adhesive member **692** such as a double-sided tape or an adhesive.

(205) A plurality of convexo-concave parts **741** may be formed in the plate support **74** in contact with the outer plate **51**, and thus, a contact rear of the adhesive or the adhesive member **741** for adhesion to the outer plate **51** is increased, so that a coupling force may be improved.

(206) The plate support **74** may be formed in all the upper frame **71**, the lower frame **72**, and the side frames **73** constituting the support frame **70**, and may be formed along the periphery of the support frame **70** to define the front surface of the support frame **70**.

(207) The plate accommodating groove **75** may be depressed at an end of the plate support **74**, and may be formed such that the bent plate part **514** bent along an opening of the outer plate **51** is inserted thereinto.

(208) Thus, in a state in which the outer plate **51** adheres to the upper frame **71**, the bent plate part **514** may be inserted into the plate accommodating groove **75**. Further, the bent plate part **514** may be in contact with a peripheral end of the transparent assembly **60** while being inserted into the plate accommodating groove **75**. Thus, when viewed from the front side, the outer plate **51** and the front surface of the transparent panel assembly **60** may be in close contact with each other without a gap therebetween.

(209) Guide ribs **751** may be formed inside the plate accommodating groove **75**. The guide ribs **751** may allow the bent plate part **514** inserted into the plate accommodating groove **75** to be in close contact with the transparent panel assembly **60**, and may guide the bent plate part **514** such that the bent plate part **514** is maintained at an accurate position while being inserted into the plate accommodating groove **75**.

(210) The guide ribs **751** may protrude to be in contact with an inner surface of the bent plate part **514**, and may extend in a direction perpendicular to an extending direction of the bent plate part **514**. The plurality of guide ribs **751** may be arranged to be adjacent to each other, and may be formed at a specific interval to entirely support a periphery of the bent plate part **514**.

(211) The guide ribs **751** may extend from one side of the inner surface of the plate accommodating groove **75** to the bottom surface of the plate accommodating groove **75**. Further, the guide ribs **751** may form inclined parts **751a** having a slope to protrude more and more from a point close to the plate support **74**. Thus, when the bent plate part **514** is inserted into the plate accommodating groove **75**, the bent plate part **514** may be inserted along the inclined parts **751a**.

(212) Further, vertical parts **751b** are formed at ends of the inclined parts **751a**, and the vertical parts **751b** may be in contact with the inner surface of the bent plate part **514** to support the bent plate part **514**. Thus, in a state in which the bent plate part **514** is completely inserted into the plate accommodating groove **75**, the bent plate part **514** may be supported by the vertical parts **751b**.

(213) Thus, while the bent plate part **514** is inserted into the plate accommodating groove **75**, the bent plate part **514** is inserted into the plate accommodating groove **75** while moving along the inclined parts **751a**, and at the same time, moves toward the end of the front panel **61**.

(214) Further, when the bent plate part **514** is completely inserted into the plate accommodating groove **75**, the bent plate part **514** may be moved to a location in contact with the front panel **61** by the vertical parts **751b**, and the bent plate part **514** may be supported while being pressed. Thus, the bent plate part **514** inserted into the plate accommodating groove **75** may be maintained in a fixed state, and a state in which the bent plate part **514** is in contact with or close to the end of the front panel **61** may be maintained.

(215) Meanwhile, restraint bosses **752** caught and restrained by one side of the bent plate part **514** may be formed inside the plate accommodating groove **75**, and when the bent plate part **514** is mounted, restrainers **514b** configured to guide the outer plate **51** such that the outer plate **51** may be

mounted at an exact position may be further formed.

(216) The panel support **76** may be formed more inward than the plate accommodating groove **75**. The panel support **76**, which is adapted to support the rear surface of the front panel **61**, defining the front surface of the transparent panel assembly **60**, may be located behind the plate support **74** and may be stepped with respect to the plate support **74**. At this time, the height difference between the panel support **76** and the plate support **74** may correspond to the thickness of the front panel **61**.

(217) Thus, in a state in which the transparent panel assembly **60** is seated on the support frame **70**, a step or a gap is not formed on the front surface of the sub-door **50**. That is, an outer end of the transparent panel assembly **60** and an end of the plate opening **511** of the outer plate **51** may be in contact with each other, and the front surface of the transparent panel assembly **60** and the front surface of the outer plate **51** are located on the same plane, so that the entire front surface of the sub-door **50** is not stepped so as to have a sense of unity. Further, the panel support **76** may be formed along the side frames **73** and the lower frame **72** except for the upper frame **71**.

(218) Meanwhile, the heater accommodating groove **761** may be formed in the panel support **76**, and the heater **532** may be accommodated inside the heat accommodating groove **761**. The heater **532** may heat the rear surface of the front panel **61**, particularly, the rear surface of the front panel **61**, which protrudes to the outside of the outer spacer **67**.

(219) In detail, the heater accommodating groove **761** may be formed on the panel support **76**. The heater accommodating groove **761**, which prevents dew condensation by heating the periphery of the transparent panel assembly **60** in contact with the panel support **76**, may be formed along the panel support **76**.

(220) The heater **532**, which is adapted to heat the periphery of the front panel **61** vulnerable to insulation, prevents dew condensation from being generated in the periphery of the front panel **61**. The heater **532** may be positioned on the vertical line of the gasket **503** inside the bezel **611**. Thus, the position in which the heater **532** is installed is an area in which a distance between the door liner **56** and the front panel **61** is close, and is relatively vulnerable to insulation. Thus, the heater **532** is arranged at the corresponding position to prevent dew condensation from being generated on the front surface of the front panel **61**. Further, the periphery of the front panel **61**, that is, the front protrusion **613**, exists between an area which is located inside the sub-door **50** and is filled with the insulator **513** and the heat insulating layer formed in the transparent panel assembly **60**, and thus is a portion in which substantially not heat insulation is provided. Thus, the periphery of the front panel **61** may be vulnerable to insulation, and the corresponding region is heated by the heater **523** so that dew condensation may not be generated in the periphery of the front panel **61**.

(221) Further, when cold air which may be transferred by the outer plate **51** is transferred to the front panel **61** due to an operation of the heater **532**, the end of the front panel **61** is heated so that dew condensation may be prevented from being generated at the end of the front panel **61**. The heater **532** may be located in the bezel **611** to heat the portion vulnerable to insulation even without being exposed to the outside so as to effectively prevent dew condensation.

(222) The heater accommodating groove **761** may be formed to have a shape corresponding to the heater **532**, and completely accommodates the heater **532**, so that when the front panel **61** is mounted, the rear surface of the front panel **61** is seated on the panel support **76**, and at this time, the heater **532** is in contact with the rear surface of the front panel **61**.

(223) In detail, when the transparent panel assembly **60** is mounted, the periphery of the front panel **61** is in contact with and seated on the panel support **76**. Further, the heater **532** mounted on the heater accommodating groove **761** may be located adjacent to the outer spacer **67**, and thus, may heat the periphery of the front panel **61**.

(224) At this time, it is preferable that the heater **532** is arranged in a region of the bezel **611** of the front panel **61**, and thus, when the transparent panel assembly **60** is mounted, the heater **532** in contact with the front panel **61** is not exposed to the outside.

(225) Meanwhile, in a state in which the heater **532** is mounted on the heater accommodating

groove **761**, an aluminum tape may be attached to shield the heater accommodating groove **761**. The aluminum tape may maintain a state in which the heater **532** is fixed and mounted to the heater accommodating groove **761**, and heat generated by the heater **532** is uniformly transferred to the periphery of the front panel **61**.

(226) The heater **532** may be formed to have a wire shape, and a generally-used sheath heater may be used as the heater **532**. The heater **532** may have a diameter at which the heater **532** may be inserted into the heater accommodating groove **761**, and may be arranged along a periphery of the frame opening **701**.

(227) Meanwhile, a vertically bent blocking part **77** may be formed at an end of the panel support **76**. The blocking part **77** may prevent foam liquid for forming the insulator **531** from being introduced toward the transparent panel assembly **60**. Further, the blocking part **77** is coupled to the outer spacer **67** through the coupling members **78** to fix the transparent panel assembly **60**.

(228) FIG. **17** is a view illustrating a coupling state of part A of FIG. **16**. Further, FIG. **18** is a sectional view taken along line **18-18'** of FIG. **17**. Further, FIG. **19** is a partial perspective view illustrating a side frame constituting the support frame. Further, FIG. **20** is a partial perspective view illustrating a lower frame constituting the support frame.

(229) The front frame has a structure in which opposite ends of the upper frame **71** and opposite ends of the lower frame **72** are coupled to opposite ends of the side frames **73**. Coupling structures thereof are identical to each other, and only locations thereof is different from each other. Thus, hereinafter, a description will be made with reference to part A of FIG. **16** in the support frame **70**.

(230) As illustrated, a lower end of the side frame **73** may be coupled to an upper end of the lower frame **72**. To achieve this, frame coupling bosses **731** may be formed at the lower end of the side frame **73**, and frame coupling grooves **721** may be formed at the upper end of the lower frame **72**, which corresponds thereto.

(231) In more detail, the frame coupling bosses **731** may protrude from the lower end of the side frame **73**, and may extend from opposite sides of the plate accommodating groove **75**, the panel support **76**, and an end of the heater accommodating groove **761**.

(232) Further, the frame coupling grooves **721** may define predetermined spaces in which the frame coupling bosses **731** may be accommodated, and may be formed in the plate accommodating groove **75**, the panel support **76**, and the heater accommodating groove **761** on the lower frame **72**.

(233) Meanwhile, a frame catching boss **731a** may be formed in one of the frame coupling bosses **731**. Further, a frame catching groove **721a** into which the frame catching boss **731a** may be inserted may be formed in one of the frame coupling grooves **721**. In a state in which the frame coupling bosses **731** and the frame coupling grooves **721** are coupled to each other, the frame catching boss **731a** may be coupled to the frame catching groove **721a**, and thus, a state in which the side frame **73** and the lower frame **72** are completely coupled to each other may be maintained.

(234) The frame coupling bosses **731** and the frame coupling grooves **721** are formed along the plate accommodating groove **75**, the panel support **76**, and the heater accommodating groove **761**, and at least portions of the frame coupling bosses **731** and the frame coupling grooves **721** are bent or extend to be perpendicular to each other, so that even when a torsional moment or a local load is applied to the support frame **70** in a state in which the frame coupling bosses **731** and the frame coupling grooves **721** are coupled to each other, a stable coupling state of the frame coupling bosses **731** and the frame coupling grooves **721** may be maintained.

(235) Further, when the side frame **73** and the lower frame **72** are coupled to each other, parting lines **L1**, **L2**, and **L3** of the ends in contact with each other may be arranged to be offset from each other. That is, the parting line **L1** at a position where portions of the plate support **74** which belong to the side frame **73** and the lower frame **72** are in contact with each other, the parting line **L2** at a position where portions of the plate accommodating groove **75** which belong to the side frame **73** and the lower frame **72** are in contact with each other, the parting line **L3** at a position where portions of the panel support **76** and the blocking part **77** which belong to the side frame **73** and the

lower frame **72** are in contact with each other may be offset from each other or may be stepped with respect to each other.

(236) Thus, when foam liquid is injected into the sub-door **50** to form the insulator **531**, the foam liquid permeates along the parting lines **L1**, **L2**, and **L3**, so that the transparent panel assembly **60** may be prevented from being polluted. That is, even when the foam liquid injected into the sub-door **50** flows along the parting line **L1** of the plate support **74**, it is difficult to introduce the foam liquid along the parting lines **L2** and **L3** of the plate accommodating groove **75** and the panel support **76**, which are arranged to be offset from each other. Thus, finally, the foam liquid may be prevented from being introduced toward the transparent panel assembly **60**.

(237) Further, the blocking part **77** may be bent to be perpendicular to an inner end of the panel support **76**, and may extend to the rear side in which the door liner **56** is located. The blocking part **77** extends from a position away from the periphery of the transparent panel assembly **60** to a position adjacent to the rear panel **65** or the door liner **56**, to prevent the foam liquid from being introduced toward the transparent panel assembly **60**.

(238) A reinforcement rib **771** may be formed at a lower end of the blocking part **77**, and deformation or damage of the blocking part **77** coupled to the transparent panel assembly **60** is prevented by the reinforcement rib **771**. Further, coupling holes **772** passing through the coupling members **78** may be formed in the blocking part **77**.

(239) FIG. **21** is a cutaway perspective view illustrating a state in which an outer plate and the support frame are coupled to each other according to the first embodiment of the present disclosure. Further, FIG. **22** is an exploded cutaway perspective view illustrating a coupling structure of the outer plate and the support frame.

(240) A coupling structure of the support frame **70** and the outer plate **51** will be described in more detail with reference to the drawings. The bent plate part **514** may be bent along the plate opening **511** at the center of the outer plate **51**.

(241) The support frame **70** may be mounted on the rear surface of the outer plate **51**. The support frame **70** may be arranged along the periphery of the plate opening **511**.

(242) The side frames **73** may be arranged at opposite left and right ends of the plate opening **511**. At this time, the bent plate part **514** may be inserted into the plate accommodating groove **75**.

(243) Meanwhile, guide ribs **751** including the vertical parts **751b** and the inclined parts **751a** may be formed inside the plate accommodating groove **75**. Thus, while the bent plate part **514** is inserted into the plate accommodating groove **75**, the bent plate part **514** may be inserted while moving along the inclined parts **751a**, and the inner surface of the bent plate part **514** may be supported by the vertical parts **751b**.

(244) The bent plate part **514** may be guided toward the inside of the plate opening **511** by the guide ribs **751**, and may maintain a position thereof in a state in which the bent plate part **514** is completely inserted into the plate accommodating groove **75**. At this time, the guide ribs **751** may support the bent plate part **514** in a manner to slightly press the bent plate part **514** from the inner side, and may prevent separation or flow of the outer plate **51**.

(245) Thus, as illustrated in FIG. **21**, in a state in which the transparent panel assembly **60** is mounted, the bent plate part **514** is located inside the plate accommodating groove **75**, and may be maintained to be in close contact with the outer end of the front panel **61**. Due to such a structure, an interval or gap between the transparent panel assembly **60** and the outer plate **51** on the front surface of the sub-door **50** cannot be virtually seen, and a boundary of the transparent panel assembly **60** and a boundary of the outer plate **51** is completely in close contact with each other, so that the entire outer appearance of the front surface of the sub-door **50** may have a sense of unity.

(246) Meanwhile, guide insertion parts **514a** may be formed on one side of the bent plate part **514** such that the outer plate **51** may be mounted on the support frame **70** at an accurate position. The guide insertion parts **514a** may be formed at an end of the bent plate part **514** so as to have a predetermined width, and may pass through the support frame **70**.

(247) Further, insertion guide holes **753** through which the guide insertion parts **514a** pass may be formed in the support frame **70**. The insertion guide holes **753** may be formed on the bottom surface of the plate accommodating groove **75**, and may have a size allowing the guide insertion parts **514a** to pass therethrough.

(248) Thus, when the outer plate **51** and the support frame **70** are coupled to each other, the outer plate **51** and the support frame **70** may be aligned with each other such that the guide insertion parts **514a** may pass through the insertion guide holes **753**, and the bent plate part **514** may be arranged inside the plate accommodating groove **75** at an accurate position.

(249) Meanwhile, when the bent plate part **514** is inserted into the plate accommodating groove **75** at an accurate position, the restraint bosses **752** formed inside the plate accommodating groove **75** may be coupled to the restrainers **514b** formed in the bent plate part **514**. In a state in which the bent plate part **514** is completely inserted into and fixed to the plate accommodating groove **75**, the restraint bosses **752** and the restrainers **514b** are coupled to each other, so that the bent plate part **514** may be maintained in an inserted state.

(250) The plurality of guide insertion parts **514a** and the plurality of restrainers **514b** may be formed in the bent plate part **514** at a specific interval. Further, the plurality of guide insertion parts **514a** and the plurality of restrainers **514b** may be formed throughout the bent plate part **514**.

(251) In this state, an adhesive or an adhesive member are applied to the plate support **74**, so that a state in which the plate support **74** is fixed and mounted to the rear surface of the outer plate **51** may be maintained. Thus, even in a situation in which the foam liquid is injected into the sub-door **50**, a position at which the support frame **70** is fixed and mounted onto the outer plate **51** may be maintained.

(252) Meanwhile, in a state in which the support frame **70** is mounted on the outer plate **51**, the transparent panel assembly **60** may be inserted and mounted from the front side to the rear side of the plate opening **511**. At this time, in a state in which the rear panel **65** having a narrow width is firstly inserted and the transparent panel assembly **60** is inserted, the rear surface of the front panel **61** may be seated on the panel support **76**.

(253) Further, in a state in which the transparent panel assembly **60** is completely inserted and mounted, the coupling members **78** fastened while passing through the blocking part **77** may be fastened to the coupling holes **672a** of the outer spacer **67**. The periphery of the transparent panel assembly **60** may be coupled to the blocking part **77** by the plurality of coupling members **78**, and the transparent panel assembly **60** may be fixed and mounted.

(254) Thus, the transparent panel assembly **60** may be firmly mounted even in a state in which an adhesive structure of the periphery of the front panel **61** and the panel support **76** is not provided, and may be maintained in a stable mounted state even when an impact is applied thereto while the sub-door **50** is opened/closed.

(255) Due to such a structure, when a problem occurs in the transparent panel assembly **60**, and thus a follow-up service is required, the transparent panel assembly **60** may be easily disassembled. Further, when the transparent panel assembly **60** is disassembled, an adhesive or an adhesive member is not applied to the bezel **611** on the periphery of the front panel **61**. Thus, the transparent panel assembly **60** is easily separated, and the bezel **611** is prevented from being damaged by the adhesive or the adhesive member as well. Thus, the follow-up service is easily performed, and the not-damaged transparent panel assembly **60** having a high price may be reused after the follow-up service.

(256) FIG. **23** is a cutaway perspective view taken along line **23-23'** of FIG. **5**. Further, FIG. **24** is a sectional view taken along line **24-24'** of FIG. **5**.

(257) As illustrated in the drawings, in a state in which the outer plate **51** and the transparent panel assembly **60** are mounted on the support frame **70**, the transparent panel assembly **60** may be fixed and mounted onto the support frame **70** through the coupling member **78**. Further, the door liner **56** is coupled, and the door lights **57** and the gasket **503** are mounted, so that the sub-door **50** is

assembled.

(258) The insulator **531** may be filled inside the assembled sub-door **50**, and the insulator **531** is filled in the outer side of the transparent panel assembly **60** to insulate a peripheral space of the sub-door **50**. Further, between the front panel **61** and the rear panel **65** of the transparent panel assembly **60**, an insulation panel **69** is provided or a sealed insulation layer is formed, so that the front panel **61** and the rear panel **65** may be insulated from each other. Thus, the insulation may be achieved throughout the entire surface of the sub-door **50**.

(259) Meanwhile, the foam liquid is injected into the sub-door **50**, the foam liquid may be prevented from being introduced toward the periphery of the transparent panel assembly **60**, by the blocking part **77**. To achieve this, the blocking part **77** may extend rearward from the end of the panel support **76**, and may extend to a position that is adjacent to the door liner **56** or the rear panel **65**. Further, if necessary, a shielding member **79** may be attached to the blocking part **77** and the door liner **56** or the rear panel **65**.

(260) The shielding member **79** is formed of an attachable material such as a tape, to completely block a gap between the shielding member **79** and the door liner **56** or the rear panel **65**. Thus, the foam liquid filled inside the sub-door **50** may be completely prevented from being introduced toward the transparent panel assembly **60**.

(261) Due to the prevention of the introduction of the foam liquid by the shielding member **79**, the foam liquid may be prevented from being polluted or being stained with the display cables **605** guided along the peripheral surface of the transparent panel assembly **60**, that is, an outer surface of the sealant **608**. That is, the display cables **605** may be located between the blocking part **77** and the sealant **608**. Thus, even when the foam liquid is injected into the sub-door **50**, the foam liquid is prevented from being introduced toward the display cables **605** by the blocking part **77**. Further, the foam liquid is not stained with the display cables **605**, and thus, even when the transparent panel assembly **60** is replaced or is separated for the follow-up service, the display cables **605** may be reused without being damaged.

(262) Hereinafter, lighting states of the display lights and the door lights will be described in more detail with reference to the accompanying drawings.

(263) FIG. **25** is a cross sectional view illustrating the main door and the sub-door. Further, FIG. **26** is a longitudinal sectional view illustrating the main door and the sub-door. Further, FIG. **27** is an enlarged view illustrating part B of FIG. **26**. Further, FIG. **28** illustrates a state in which an interior of the refrigerator may be seen through the transparent panel assembly. Further, FIG. **29** illustrates a state in which a screen is output through the transparent panel assembly.

(264) As illustrated in the drawings, in a state in which the locking member **593** of the opening device **59** is inserted into a latch hole **421**, a state in which the sub-door **50** is closed is maintained. In this state, a state in which the door light **57** is turned off is maintained. An opened/closed state of the sub-door **50** may be detected through a separately provided door switch.

(265) As illustrated in FIG. **1**, in a state in which the door lights **57** are turned off, the rear space of the sub-door **50** becomes dark, so that the interior of the refrigerator **1** cannot be seen through the see-through part **21**. Thus, when there is no separate manipulation in a state in which the sub-door **50** is closed, the door lights **57** are continuously turned off, and thus, the interior of the refrigerator **1** cannot be seen through the see-through part **21**.

(266) In this state, the user touches the front panel **51** to switch off the door lights **57**. When the door lights **57** are turned on, light beams irradiated by lighting modules **575** are irradiated from opposite left and right sides to the central side of the rear side of the rear panel **65** to face each other.

(267) The door lights **57** may extend an upper end to a lower end of the rear panel **65**. That is, the light beams irradiated by the door lights **57** may illuminate the entire rear region of the rear panel **65** on the opposite left and right sides of the rear panel **65**.

(268) At this time, when the display lights **86** are turned on together, light beams may be irradiated

from the upper side and the lower side by the display lights **68**, and the light beams may be irradiated from the left side and the right side by the door lights **57**. As a result, the light beams may be irradiated from all the upper, lower, left, and right sides of the see-through part **21**, and a region of the see-through part **21** may be illuminated in the maximum brightness.

(269) The door lights **57** irradiates the light beams in a direction in which the light beams face each other, while being adjacent to the rear panel **65**. The light beams irradiated by the door lights **57** may illuminate the internal space of the storage case **43**, and may illuminate the front side via the rear panel **65** as well. Thus, as illustrated in FIG. **28**, the door lights **57** may serve as lights configured to illuminate a space inside the refrigerator **1**, which is seen through the see-through part **21**, and at the same time, may serve as auxiliary backlights through which the display **62** may be seen more clearly.

(270) That is, in a state in which the screen is output through the display **62**, the space inside the refrigerator **1**, that is, a space behind the sub-door **50**, may be selectively seen through the see-through part **21**. To allow the space behind the sub-door **50** to be seen through the see-through part **21**, the door lights **57** may be turned on.

(271) Of course, various representations may be achieved through a combination of ON/OFF states of the display lights **68** and the door lights **57** according to a degree to which the inside of the storage case **43** is visualized through the see-through part **21**.

(272) Further, when the user manipulates the front panel **61** on the front surface of the refrigerator **1**, the display lights **68** are turned on, the display **62** is turned on, and thus, the transparent panel assembly **60** may output the screen, as illustrated in FIG. **29**. At this time, the manipulation of the front panel **61** may correspond to input of any one of a specific position, the number of times of touches, and a pattern. Of course, if necessary, the manipulation by the user may be detected using a separate physical button or a sensor.

(273) The display **62** may output the screen for displaying a state of the refrigerator **1** and performing manipulation, and may also output various screens for performing the Internet, outputting an image, performing output using an external input device, and displaying information on received food.

(274) In detail, the display lights **68** arranged at an upper end and a lower end of the light guide plate **64** may be turned on together with the display **62** by the manipulation by the user. The display lights **68** are turned on, and thus the light guide plate **64** diffusely reflects and diffuses light of the display lights **68**, so that the light may be irradiated toward the display **62** on the front side in a wholly uniform brightness.

(275) The light is irradiated from the rear side of the display **62** toward the display **62** by the light guide plate **64**, and at the same time, the screen is output based on image information input from the display **62**. Thus, the user may identify the clearly output screen through the see-through part **21**.

(276) Meanwhile, the operation of the display **62** and the operations of the door lights **57** may be controlled by the PCBs **602**, **603**, and **604** such as the T-CON board **602** or the docking PCB **604** above the sub-door **50**. Further, these PCBs **602**, **603**, and **604** may be arranged on the rear space of the sub-door **50**, which is partitioned by the barrier **711** defining the upper end of the support frame **70**. Further, the insulator **531a** may be filled in a front space of the sub-door **50**, which is partitioned by the barrier **711**, and thus dew condensation may be prevented from being generated on an upper side of the front surface of the sub-door **50**.

(277) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(278) In a second embodiment of the present disclosure, the support frame configured to support the outer case and the transparent panel assembly such that ends of the outer case and the transparent panel assembly are in contact with each other is provided, and the support frame is coupled to any one of the plurality of spacers.

(279) In description of the second embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted. Further, not-illustrated reference numerals will be the same as the configurations of the drawings in the above-described embodiments.

(280) FIG. 30 is a sectional view illustrating a door according to a second embodiment of the present disclosure.

(281) Referring to the drawing, an outer peripheral shape of the door 50 may be defined by the bent outer plate 51 formed of metal. The outer plate 51 may define the front surface and a periphery of the side surfaces of the door 50. Further, the door liner 56 defining the rear surface of the door 50 is coupled to the outer plate 51, and the transparent panel assembly 60 is provided in openings of the outer plate 51 and the door liner 56, so that the interior of the refrigerator 1 may be selectively seen.

(282) The transparent panel assembly 60 may include the front panel 61 defining the front surface thereof, the rear panel 65 defining the rear surface thereof, and the insulation panel 69 between the front panel 61 and the rear panel 65. A metal deposition layer or a film layer through which light is selectively passes may be formed on the rear surface of the front panel 61, and thus the interior of the refrigerator 1 may be selectively visualized according to whether the door lights 57 or a lamp in the refrigerator 1 is turned on or off.

(283) The display 62 may be provided in the transparent panel assembly 60 as in the above-described first embodiment, and at this time, the light guide plate 64 may be further provided. Further, the light guide plate 64 may be provided instead of the insulation panel 69.

(284) The front panel 61, the insulation panel 69, and the rear panel 65 may be arranged at a set interval by a third spacer 661 and a fourth spacer 662, and sealed insulation spaces may be formed between the panels.

(285) Further, a spacer protrusion 661a may be formed on one side of the third spacer 661, and the coupling members 78 such as screws may be fastened to the spacer protrusion 661a. The coupling members 78 may be fastened through a support frame 170 configured to support the outer plate 51 and the transparent panel assembly 60, and thus the transparent panel assembly 60 may be fixed and mounted onto the support frame 170.

(286) The sealant 608 may be applied to spaces on opposite sides with respect to the spacer protrusion 661a. The sealant 608 may be applied along the periphery of the transparent panel assembly 60, and may protrude to the same height as that of the spacer protrusion 661a.

(287) Meanwhile, the support frame 170 may include a plate support 171 configured to support the outer plate 51, a panel support 172 configured to support the periphery of the front panel 61, and a blocking part 173 configured to prevent the foam liquid from permeating along the peripheral surface of the transparent panel assembly 60.

(288) The plate support 171 may adhere to the rear surface of the outer plate 51 by an adhesive member 171a. At this time, an end of the plate support 171 may be situated at a position corresponding to an end defining the opening of the outer plate 51.

(289) Further, the panel support 172 is stepped with respect to the plate support 171, and thus the periphery of the front panel 61 further protruding outward may be seated on the panel support 172. At this time, the panel support 172 may be stepped with respect to the plate support 171 by the thickness of the front panel 61.

(290) Thus, in a state in which the transparent panel assembly 60 is mounted, the outer plate 51 and the front surface of the front panel 61 may be located at the same height and may be located on the same plane. Further, the end of the outer plate 51 and the outer end of the front panel 61 are in contact with each other, and thus when viewed from the outside, a gap between the outer plate 51 and the front panel 61 cannot be viewed.

(291) Meanwhile, a heater accommodating groove on which the heater 532 is mounted may be formed in the panel support 172 as in the above-described first embodiment.

(292) The blocking part **173** may extend from the panel support **172**, and may vertically extend from the end of the panel support **172** to the door liner **56**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the blocking part **173** may extend to be in contact with the door liner **56**, and when the blocking part **173** is spaced apart from the door liner **56**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at an end of the blocking part **173** as in the above-described first embodiment.

(293) Meanwhile, the coupling members **78** such as screws may be fastened to the blocking part **173**, and the coupling members **78** may pass through the blocking part **173** to be coupled to the spacer protrusion **661a**. Thus, the transparent panel assembly **60** may adhere to the support frame **170** without a separate configuration such as adhesive.

(294) That is, the transparent panel assembly **60** may be fixed and mounted onto the support frame **170** by the blocking part **173**. Thus, the transparent panel assembly **60** may be firmly fixed, the transparent panel assembly **60** may be separated, and serviceability may be improved. Further, a separate configuration for adhesion is not provided in the bezel **611** at the periphery of the front panel **61**, so that even when the transparent panel assembly **60** is separated, the bezel **611** may be prevented from being damaged.

(295) Further, as the foam liquid is prevented from being introduced by the blocking part **173**, the transparent panel assembly **60** may be separated, and serviceability may be improved. The cables connected to the electric components for operating the transparent panel assembly **60** are arranged along the periphery of the transparent panel assembly **60**, so that the cables may be prevented from being polluted or damaged by the foam liquid.

(296) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(297) In a third embodiment, a single spacer configured to support the outer case and a plurality of panels of the transparent panel assembly is provided, and is coupled to the support frame configured to support the outer plate and the front panel, by the coupling members.

(298) In description of the third embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(299) FIG. **31** is a sectional view illustrating a door according to a third embodiment of the present disclosure.

(300) Referring to the drawing, the door **50** may be configured by the outer plate **51**, the door liner **56**, and the transparent panel assembly **60**.

(301) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**. A metal deposition layer or a film layer through which light is selectively passes may be formed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(302) The front panel **61**, the insulation panel **69**, and the rear panel **65** may be arranged at a set interval by a fifth spacer **663**, and sealed insulation spaces may be formed between the panels.

(303) The fifth spacer **663**, which is a single configuration, configures the outermost side of the transparent panel assembly **6**, and allows the insulation panel **69** to be fixed between the front panel **61** and the rear panel **65**.

(304) In detail, a depressed panel accommodating groove **663a** accommodating an end of the insulation panel **69** may be formed at the center of the inner surface of the fifth spacer **663**. In a state in which the insulation panel **69** is mounted inside the panel accommodating groove **663a**, the front panel **61** and the rear panel **65** are mounted on the front surface of the rear surface of the fifth spacer **663**, so that the insulation panel **69**, the front panel **61**, and the rear panel **65** may be

arranged at a set interval, and a sealed insulation space may be formed.

(305) Meanwhile, a spacer protrusion **663b** may be formed on an outer surface of the fifth spacer **663**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **663b**. The coupling members **78** may be fastened through the support frame **170** configured to support the outer plate **51** and the transparent panel assembly **60**, and thus the transparent panel assembly **60** may be fixed and mounted onto the support frame **170**.

(306) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **663b**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **663b**.

(307) Meanwhile, the support frame **170** may include a plate support **171** configured to support the outer plate **51**, a panel support **172** configured to support the periphery of the front panel **61**, and a blocking part **173** configured to prevent the foam liquid from permeating along the peripheral surface of the transparent panel assembly **60**.

(308) The front surface of the outer plate **51** and the front surface of the front panel **61** may be located on the same plane by the support frame **170** so as not to be stepped with respect to each other. Further, the end defining the opening of the outer plate **51** is in contact with the outer end of the front panel **61**, so that a gap between the outer plate **51** and the front panel **61** is not exposed.

(309) Further, the transparent panel assembly **60** may be fixed and mounted onto the support frame **170** by the blocking part **173**. Thus, the transparent panel assembly **60** may be firmly fixed, the transparent panel assembly **60** may be separated, and serviceability may be improved. Further, a separate configuration for adhesion is not provided in the bezel **611** at the periphery of the front panel **61**, so that even when the transparent panel assembly **60** is separated, the bezel **611** may be prevented from being damaged.

(310) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(311) A fourth embodiment of the present disclosure is characterized in that the outer plate and the front panel may be mounted to be in contact with each other by the support frame configured to support the outer plate and the front panel.

(312) In description of the fourth embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(313) FIG. **32** is a sectional view illustrating a door according to a fourth embodiment of the present disclosure.

(314) Referring to the drawing, the door **50** may be configured by the outer plate **51**, the door liner **56**, and the transparent panel assembly **60**.

(315) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**. A metal deposition layer or a film layer through which light is selectively passes may be formed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(316) Meanwhile, a sixth spacer **664** may be provided between the front panel **61** and the insulation panel **69**, and a seventh spacer **665** may be provided between the insulation panel **69** and the rear panel **65**. The front panel **61**, the insulation panel **69**, and the rear panel **65** may be arranged at a specific interval by the sixth spacer **664** and the seventh spacer **665**.

(317) Further, an outer spacer **667** may be provided outside the insulation panel **69**. The outer spacer **667** connects the front panel **61** and the rear panel **65** between the front panel **61** and the rear panel **65**, and the insulation panel **69**, the sixth spacer **664**, and the seventh spacer **665** may be provided in an internal sealed space. Thus, a space between the front panel **61** and the rear panel **65** is sealed due to the sealing of the outer spacer **667**, to form an insulation layer.

(318) Meanwhile, a spacer protrusion **667a** may be formed on an outer surface of the outer spacer **667**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **667a**. The coupling members **78** may be fastened through the support frame **170** configured to support the outer plate **51** and the transparent panel assembly **60**, and thus the transparent panel assembly **60** may be fixed and mounted onto the support frame **170**.

(319) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **667a**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **667a**.

(320) Meanwhile, the support frame **170** may include a plate support **171** configured to support the outer plate **51**, a panel support **172** configured to support the periphery of the front panel **61**, and a blocking part **173** configured to prevent the foam liquid from permeating along the peripheral surface of the transparent panel assembly **60**.

(321) The front surface of the outer plate **51** and the front surface of the front panel **61** may be located on the same plane by the support frame **170** so as not to be stepped with respect to each other. Further, the end defining the opening of the outer plate **51** is in contact with the outer end of the front panel **61**, so that a gap between the outer plate **51** and the front panel **61** is not exposed.

(322) Further, the transparent panel assembly **60** may be fixed and mounted onto the support frame **170** by the blocking part **173**. Thus, the transparent panel assembly **60** may be firmly fixed, the transparent panel assembly **60** may be separated, and serviceability may be improved. Further, a separate configuration for adhesion is not provided in the bezel **611** at the periphery of the front panel **61**, so that even when the transparent panel assembly **60** is separated, the bezel **611** may be prevented from being damaged.

(323) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(324) A fifth embodiment of the present disclosure is characterized in that the transparent panel assembly may be supported by the outer plate, and by the blocking part formed in the outer plate, the transparent panel assembly may be fixed, and permeation of the foam liquid may be prevented.

(325) In description of the fifth embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(326) FIG. **33** is a sectional view illustrating a door according to a fifth embodiment of the present disclosure.

(327) Referring to the drawing, an outer peripheral shape of the door **50** may be defined by the bent outer plate **51** formed of metal. The outer plate **51** may define the front surface and a periphery of the side surfaces of the door **50**. Further, the door liner **56** defining the rear surface of the door **50** is coupled to the outer plate **51**, and the transparent panel assembly **60** is provided in openings of the outer plate **51** and the door liner **56**, so that the interior of the refrigerator **1** may be selectively seen.

(328) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof and configured to selectively visualize the interior of the refrigerator **1**, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**.

(329) The front panel **61**, the insulation panel **69**, and the rear panel **65** may be arranged at a set interval by a third spacer **661** and a fourth spacer **662**, and sealed insulation spaces may be formed between the panels.

(330) Further, a spacer protrusion **661a** may be formed on one side of the third spacer **661**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **661a**. The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **661a**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **661a**.

(331) Meanwhile, an opening into which the transparent panel assembly **60** is inserted from the front side may be formed on the front surface part **512** defining the front surface of the outer plate **51**. Further, a mounting part **515** and the blocking part **516** may be formed at an inner end of the front surface part **512**, and an inner surface of the opening passing through the door **50** may be formed by the mounting part **515** and the blocking part **516**.

(332) In detail, the mounting part **515** may be inward stepped with respect to an end of the front surface part **512**. At this time, the mounting part **515** may be stepped with respect to the front surface part **512** by the thickness of the front panel **61**.

(333) Thus, in a state in which the transparent panel assembly **60** is mounted, the outer plate **51** and the front surface of the front panel **61** may be located at the same height and may be located on the same plane. Further, the end of the front surface part **512** and the outer end of the front panel **61** are in contact with each other, and thus when viewed from the outside, a gap between the outer plate **51** and the front panel **61** cannot be viewed.

(334) Meanwhile, the heater **532** may be mounted on the rear surface of the mounting part **515** as in the above-described first embodiment, and may heat the periphery of the front panel **61**, thereby preventing dew condensation.

(335) The blocking part **516** may vertically extend from the mounting part **515** to the door liner **56**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the blocking part **516** may extend to be in contact with the door liner **56**, and when the blocking part **516** is spaced apart from the door liner **56**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at an end of the blocking part **173** as in the above-described first embodiment.

(336) Meanwhile, the coupling members **78** such as screws may be fastened to the blocking part **516**, and the coupling members **78** may pass through the blocking part **516** to be coupled to the spacer protrusion **661a**. Thus, the transparent panel assembly **60** may adhere to the outer plate **51** without a separate configuration such as adhesive.

(337) That is, the transparent panel assembly **60** may be fixed and mounted onto the outer plate **51** by the blocking part **516**. Thus, the transparent panel assembly **60** may be firmly fixed, the transparent panel assembly **60** may be separated, and serviceability may be improved. Further, a separate configuration for adhesion is not provided in the bezel **611** at the periphery of the front panel **61**, so that even when the transparent panel assembly **60** is separated, the bezel **611** may be prevented from being damaged.

(338) Further, as the foam liquid is prevented from being introduced by the blocking part **516**, the transparent panel assembly **60** may be separated, and serviceability may be improved. The cables connected to the electric components for operating the transparent panel assembly **60** are arranged along the periphery of the transparent panel assembly **60**, so that the cables may be prevented from being polluted or damaged by the foam liquid.

(339) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(340) A sixth embodiment of the present disclosure is characterized in that the transparent panel assembly may be supported by the outer plate, and by coupling between a single-structural spacer and the blocking part formed in the outer plate, the transparent panel assembly may be fixed, and permeation of the foam liquid may be prevented.

(341) In description of the sixth embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(342) FIG. **34** is a sectional view illustrating a door according to a sixth embodiment of the present disclosure.

(343) Referring to the drawing, an outer peripheral shape of the door **50** may be defined by the bent

outer plate **51** formed of metal. The outer plate **51** may define the front surface and a periphery of the side surfaces of the door **50**. Further, the door liner **56** defining the rear surface of the door **50** is coupled to the outer plate **51**, and the transparent panel assembly **60** is provided in openings of the outer plate **51** and the door liner **56**, so that the interior of the refrigerator **1** may be selectively seen.

(344) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof and configured to selectively visualize the interior of the refrigerator **1**, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**.

(345) The front panel **61**, the insulation panel **69**, and the rear panel **65** may be arranged at a set interval by a fifth spacer **663**, and sealed insulation spaces may be formed between the panels.

(346) The fifth spacer **663**, which is a single configuration, configures the outermost side of the transparent panel assembly **60**, and allows the insulation panel **69** to be fixed between the front panel **61** and the rear panel **65**.

(347) In detail, a depressed panel accommodating groove **663a** accommodating an end of the insulation panel **69** may be formed at the center of the inner surface of the fifth spacer **663**. In a state in which the insulation panel **69** is mounted inside the panel accommodating groove **663a**, the front panel **61** and the rear panel **65** are mounted on the front surface of the rear surface of the fifth spacer **663**, so that the insulation panel **69**, the front panel **61**, and the rear panel **65** may be arranged at a set interval, and a sealed insulation space may be formed.

(348) Meanwhile, a spacer protrusion **663b** may be formed on an outer surface of the fifth spacer **663**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **663b**. The coupling members **78** may be fastened through the blocking part **516**, and thus the transparent panel assembly **60** may be fixed and mounted onto the outer plate **51**.

(349) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **663b**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **663b**.

(350) Meanwhile, the front surface part **512** defining the front surface of the outer plate **51** may be formed on the outer plate **51**, and the mounting part **515** and the blocking part **516** may be formed at an inner end of the front surface part **512** having an opening formed therein.

(351) In detail, the mounting part **515** may be inward stepped with respect to an end of the front surface part **512**. At this time, the mounting part **515** may be stepped with respect to the front surface part **512** by the thickness of the front panel **61**.

(352) Thus, in a state in which the transparent panel assembly **60** is mounted, the outer plate **51** and the front surface of the front panel **61** may be located at the same height and may be located on the same plane. Further, the end of the front surface part **512** and the outer end of the front panel **61** are in contact with each other, and thus when viewed from the outside, a gap between the outer plate **51** and the front panel **61** cannot be viewed.

(353) The blocking part **516** may vertically extend from the mounting part **515** to the door liner **56**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the blocking part **516** may extend to be in contact with the door liner **56**, and when the blocking part **516** is spaced apart from the door liner **56**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at an end of the blocking part **173** as in the above-described first embodiment.

(354) The coupling members **78** such as screws may be fastened to the blocking part **516**, and the coupling members **78** may pass through the blocking part **516** to be coupled to the spacer protrusion **663b**. Thus, the transparent panel assembly **60** may adhere to the outer plate **51** without a separate configuration such as adhesive.

(355) Meanwhile, various other embodiments in addition to the above-described embodiments may

be applied to the refrigerator according to the present disclosure.

(356) A seventh embodiment of the present disclosure is characterized in that the transparent panel assembly may be supported by the outer plate, and by coupling between double-structural spacers and the blocking part formed in the outer plate, the transparent panel assembly may be fixed, and permeation of the foam liquid may be prevented.

(357) In description of the seventh embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(358) FIG. 35 is a sectional view illustrating a door according to a seventh embodiment of the present disclosure.

(359) Referring to the drawing, an outer peripheral shape of the door **50** may be defined by the bent outer plate **51** formed of metal. The outer plate **51** may define the front surface and a periphery of the side surfaces of the door **50**. Further, the door liner **56** defining the rear surface of the door **50** is coupled to the outer plate **51**, and the transparent panel assembly **60** is provided in openings of the outer plate **51** and the door liner **56**, so that the interior of the refrigerator **1** may be selectively seen.

(360) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof and configured to selectively visualize the interior of the refrigerator **1**, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**.

(361) Meanwhile, a sixth spacer **664** may be provided between the front panel **61** and the insulation panel **69**, and a seventh spacer **665** may be provided between the insulation panel **69** and the rear panel **65**. The front panel **61**, the insulation panel **69**, and the rear panel **65** may be arranged at a specific interval by the sixth spacer **664** and the seventh spacer **665**.

(362) Further, an outer spacer **667** may be provided outside the insulation panel **69**. The outer spacer **667** connects the front panel **61** and the rear panel **65** between the front panel **61** and the rear panel **65**, and the insulation panel **69**, the sixth spacer **664**, and the seventh spacer **665** may be provided in an internal sealed space. Thus, a space between the front panel **61** and the rear panel **65** is sealed due to the sealing of the outer spacer **667**, to form an insulation layer.

(363) Meanwhile, a spacer protrusion **667a** may be formed on an outer surface of the outer spacer **667**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **667a**. The coupling members **78** may be fastened through the blocking part **516**, and thus the transparent panel assembly **60** may be fixed and mounted onto the outer plate **51**.

(364) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **667a**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **667a**.

(365) Meanwhile, the front surface part **512** defining the front surface of the outer plate **51** may be formed on the outer plate **51**, and the mounting part **515** and the blocking part **516** may be formed at an inner end of the front surface part **512** having an opening formed therein.

(366) In detail, the mounting part **515** may be inward stepped with respect to an end of the front surface part **512**. At this time, the mounting part **515** may be stepped with respect to the front surface part **512** by the thickness of the front panel **61**.

(367) Thus, in a state in which the transparent panel assembly **60** is mounted, the outer plate **51** and the front surface of the front panel **61** may be located at the same height and may be located on the same plane. Further, the end of the front surface part **512** and the outer end of the front panel **61** are in contact with each other, and thus when viewed from the outside, a gap between the outer plate **51** and the front panel **61** cannot be viewed.

(368) The blocking part **516** may vertically extend from the mounting part **515** to the door liner **56**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the

blocking part **516** may extend to be in contact with the door liner **56**, and when the blocking part **516** is spaced apart from the door liner **56**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at an end of the blocking part **173** as in the above-described first embodiment.

(369) The coupling members **78** such as screws may be fastened to the blocking part **516**. The coupling members **78** may be fastened through the blocking part **516**, and thus the transparent panel assembly **60** may be fixed and mounted onto the outer plate **51** without a separate configuration such as adhesive.

(370) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(371) An eighth embodiment of the present disclosure is characterized in that an opening is formed in the door, and the transparent panel assembly is mounted on the rear side of the opening, and is fixed and mounted by the support frame mounted on the door liner.

(372) In description of the eighth embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(373) FIG. **36** is a sectional view illustrating a door according to an eighth embodiment of the present disclosure.

(374) Referring to the drawing, an outer peripheral shape of the door **50** may be defined by the bent outer plate **51** formed of metal. The outer plate **51** may define the front surface and a periphery of the side surfaces of the door **50**. Further, the door liner **56** defining the rear surface of the door **50** is coupled to the outer plate **51**, and the transparent panel assembly **60** is provided in openings of the outer plate **51** and the door liner **56**, so that the interior of the refrigerator **1** may be selectively seen.

(375) The transparent panel assembly **60** may include the front panel **61a** defining the front surface thereof and configured to selectively visualize the interior of the refrigerator **1**, the rear panel **65a** defining the rear surface thereof, and the insulation panel **69** between the front panel **61a** and the rear panel **65a**.

(376) At this time, the front panel **61a** is formed to be smaller than the opening, and the rear panel **65a** is formed to correspond to the size of the opening, and thus may be formed to be larger than the front panel **61a**. Thus, the transparent panel assembly **60** may be mounted while being inserted from the rear side of the door **50**.

(377) The front panel **61a**, the insulation panel **69**, and the rear panel **65a** may be arranged at a set interval by the third spacer **661** and the fourth spacer **662**, and sealed insulation spaces may be formed between the panels.

(378) Further, a spacer protrusion **661a** may be formed on one side of the third spacer **661**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **661a**. The coupling members **78** may be fastened through a support frame **270** configured to support the outer plate **51** and the transparent panel assembly **60**, and thus the transparent panel assembly **60** may be fixed and mounted onto the support frame **270**.

(379) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **661a**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **661a**.

(380) Meanwhile, the support frame **270** may include a liner support **271** configured to support the door liner **56**, a panel support **272** configured to support the periphery of the rear panel **65a**, and a blocking part **273** configured to prevent the foam liquid from permeating along the peripheral surface of the transparent panel assembly **60**.

(381) The liner support **271** may adhere to the rear surface of the door liner **56** by an adhesive member. At this time, an end of the liner support **271** may be situated at a position corresponding to an end defining the opening of the door liner **56**. Further, the panel support **272** is stepped with

respect to the plate support **271**, and thus the periphery of the rear panel **65a** further protruding outward may be seated on the panel support **272**.

(382) The blocking part **273** may extend from the panel support **272**, and may vertically extend from the end of the panel support **272** to the front panel **61a**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the blocking part **273** may extend to be in contact with the front panel **61a** or the outer plate **51**, and when the blocking part **273** is spaced apart from the front panel **61a** or the outer plate **51**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at the end of the blocking part **173** as in the above-described first embodiment.

(383) Meanwhile, the coupling members **78** such as screws may be fastened to the blocking part **273**. Thus, the coupling members **78** may pass through the blocking part **273** to be coupled to the spacer protrusion **661a**. Thus, the transparent panel assembly **60** may adhere to the support frame **270** without a separate configuration such as adhesive. Thus, the transparent panel assembly **60** may be firmly fixed, and may be easily separated, and serviceability may be improved.

(384) Further, as the foam liquid is prevented from being introduced by the blocking part **273**, the transparent panel assembly **60** may be separated, and serviceability may be improved. The cables connected to the electric components for operating the transparent panel assembly **60** are arranged along the periphery of the transparent panel assembly **60**, so that the cables may be prevented from being polluted or damaged by the foam liquid.

(385) In a state in which the transparent panel assembly **60** is fixed and mounted onto the support frame **270** by the coupling members **78**, the front surface of the outer plate **51** and the front surface of the front panel **61a** may be arranged on the same plane. That is, the stepped height of the panel support **272** may be formed such that the front panel **61a** may be located to coincide with the front surface of the outer plate **51**.

(386) Further, the outer end of the front panel **61a** may be arranged to be in contact with the end of the outer plate **51**. Thus, when the front surface of the door **50** is viewed, the outer plate **51** and the front panel **61a** may be stepped with respect to each other or a gap between the outer plate **51** and the front panel **61** may not be generated.

(387) Meanwhile, various other embodiments in addition to the above-described embodiments may be applied to the refrigerator according to the present disclosure.

(388) A ninth embodiment of the present disclosure is characterized in that an opening is formed in the door, the transparent panel assembly is fixed and mounted onto the rear side of the opening, and the door liner is fixed and supported by the transparent panel assembly.

(389) In description of the ninth embodiment of the present disclosure, the same configurations according to the above-described embodiments will be designated by the same reference numerals, and detailed descriptions thereof will be omitted.

(390) FIG. **37** is a sectional view illustrating a door according to a ninth embodiment of the present disclosure.

(391) Referring to the drawing, an outer peripheral shape of the door **50** may be defined by the bent outer plate **51** formed of metal. The outer plate **51** may define the front surface and a periphery of the side surfaces of the door **50**. Further, the door liner **56** defining the rear surface of the door **50** is coupled to the outer plate **51**, and the transparent panel assembly **60** is provided in openings of the outer plate **51** and the door liner **56**, so that the interior of the refrigerator **1** may be selectively seen.

(392) The transparent panel assembly **60** may include the front panel **61a** defining the front surface thereof and configured to selectively visualize the interior of the refrigerator **1**, the rear panel **65a** defining the rear surface thereof, and the insulation panel **69** between the front panel **61a** and the rear panel **65a**.

(393) At this time, the front panel **61a** may be formed to be smaller than the opening, and the rear

panel **65a** may be formed to correspond to the opening and thus to be larger than the front panel **61a**. Thus, the transparent panel assembly **60** may be mounted while being inserted from the rear side of the door **50**.

(394) Meanwhile, the sixth spacer **664** may be provided between the front panel **61a** and the insulation panel **69**, and the seventh spacer **665** may be provided between the insulation panel **69** and the rear panel **65a**. The front panel **61a**, the insulation panel **69**, and the rear panel **65a** may be arranged at a specific interval by the sixth spacer **664** and the seventh spacer **665**.

(395) Further, the outer spacer **667** may be provided outside the insulation panel **69**. The outer spacer **667** connects the front panel **61a** and the rear panel **65a** between the front panel **61** and the rear panel **65**, and the insulation panel **69**, the sixth spacer **664**, and the seventh spacer **665** may be provided in an internal sealed space. Thus, a space between the front panel **61a** and the rear panel **65a** is sealed due to the sealing of the outer spacer **667**, to form an insulation layer.

(396) Meanwhile, the spacer protrusion **667a** may be formed on the outer surface of the outer spacer **667**, and the coupling members **78** such as screws may be fastened to the spacer protrusion **667a**. The coupling members **78** may be fastened through the blocking part **563**, and thus the transparent panel assembly **60** may be fixed and mounted onto the door liner **56**.

(397) The sealant **608** may be applied to spaces on opposite sides with respect to the spacer protrusion **667a**. The sealant **608** may be applied along the periphery of the transparent panel assembly **60**, and may protrude to the same height as that of the spacer protrusion **667a**.

(398) Meanwhile, the door liner **56** may be formed on the rear surface of the door **50**, and the gasket **503** may be mounted to the door liner **56**. Further, a mounting part **562** and a blocking part **563** on which the rear panel **65a** is seated may be formed in the door liner **56**.

(399) The mounting part **562** may be formed at an end of the door liner **56** in which the opening is formed, and may be stepped such that the outer end of the rear panel **65a** may be seated thereon. Further, when the rear panel **65a** is mounted on the mounting part **562**, the mounting part **562** may have a height at which the front surface of the front panel **61a** may be located at the same position as that of the front surface of the outer plate **51**.

(400) Further, the outer end of the front panel **61a** may be arranged to be in contact with the end of the outer plate **51**. Thus, when the front surface of the door **50** is viewed, the outer plate **51** and the front panel **61a** may be stepped with respect to each other or the gap between the outer plate **51** and the front panel **61** may not be generated.

(401) The blocking part **563** may extend from the mounting part **562**, and may vertically extend from the end of the mounting part **562** to the front panel **61a**. Thus, the foam liquid filled to form the insulator **531** formed inside the sub-door **50** may be prevented from being introduced toward the transparent panel assembly **60**. At this time, the blocking part **563** may extend to be in contact with the front panel **61a** or the outer plate **51**, and when the blocking part **563** is spaced apart from the front panel **61a** or the outer plate **51**, the shielding member **79** for preventing the introduction of the foam liquid may be provided at the end of the blocking part **173** as in the above-described first embodiment.

(402) Meanwhile, the coupling members **78** such as screws may be fastened to the blocking part **563**, and the coupling members **78** may pass through the blocking part **563** to be coupled to the spacer protrusion **667a**. Thus, the transparent panel assembly **60** may adhere to the door liner **56** without a separate configuration such as adhesive. Thus, the transparent panel assembly **60** may be firmly fixed, and may be easily separated, and serviceability may be improved.

(403) Further, the foam liquid is prevented from being introduced by the blocking part **563**, so that the transparent panel assembly **60** may be easily separated and a service may be easily performed. Further, cables connected to electric components for operating the transparent panel assembly **60** are arranged along the periphery of the transparent panel assembly **60**, so that the cables may be prevented from being polluted or damaged by the foam liquid.

(404) Meanwhile, various other embodiments in addition to the above-described embodiments may

be applied to the present disclosure.

(405) Tenth to twelfth embodiments of the present disclosure is characterized in that the doors according to the above-described embodiments may be applied to refrigerators having various structures.

(406) In the following embodiments, there is merely a difference only in the position and the size of the doors, and the structures of the doors according to the above-described embodiments may be applied. Thus, the same reference numerals will be used and the detailed descriptions thereof will be omitted.

(407) FIG. **38** is a perspective view of the sub-door when viewed from a front side. Also, FIG. **39** is an exploded perspective view of the sub-door.

(408) As illustrated in the drawings, the sub-door **50** may include an outer plate **51** defining an outer appearance of the sub-door **50**, a door liner **56** mounted to be spaced apart from the outer plate **51**, the transparent panel assembly **60** mounted on an opening of the outer plate **51** and the door liner **56**, and upper and lower cap decors **54** and **55** defining the top and bottom surfaces of the sub-door **50**. The above-described constituents may be coupled to define the whole outer appearance of the sub-door **50**.

(409) Also, a door light **57** may be provided on each of both sides of the door liner opening **561**. Also, the opening device **59** may be mounted on the door liner **56**.

(410) The transparent panel assembly **60** may be disposed between the outer plate **51** and the door liner **56**. The inner frame **70** for supporting the transparent panel assembly **60** is mounted on a periphery of the plate opening **511** of the outer plate **51**. The transparent panel assembly **60** may be fixed to and mounted on the outer plate **51** by the support frame **70**.

(411) A bezel **611** covering the coupled structure around the transparent panel assembly **60** so that predetermined light is not transmitted may be disposed around the transparent panel assembly **60**. The bezel **611** may have a black color to completely shield the inside thereof and may have a predetermined width. Thus, an area inside the bezel **611** may be defined as the see-through part **21**. Also, a portion of the support frame **70**, which supports a periphery of the transparent panel assembly **60**, may be disposed on the area of the bezel **611** and thus covered so that the inside thereof is not seen from the outside.

(412) The transparent panel assembly **60** may not include a display **62** for outputting a screen, and the transparent panel assembly **60** without the display **62** may have the same outer appearance as that of the transparent panel assembly having the display **62** only except that a screen is not outputted. Thus, the structure for fixing and supporting the transparent panel assembly **60** and the structure for preventing dew condensation from being generated on the surface of the transparent panel assembly **60** may be equally applicable.

(413) Hereinafter, the structure of the transparent panel assembly will be described in more detail.

(414) FIG. **40** is a perspective view of the transparent panel assembly according to a tenth embodiment of the present disclosure. Also, FIG. **41** is an exploded perspective view of the transparent panel assembly. Also, FIG. **42** is a cross-sectional view of the transparent panel assembly.

(415) As illustrated in the drawings, the transparent panel assembly **60** may be constituted by front and rear panels **61** and **65** defining at least front and rear surfaces and a spacer **67** connecting the front panel **61** to the rear panel **65**. Also, additional panel and spacer may be further provided in an inner space defined by the spacer **67**. Also, the inner space defined by the spacer and the panels may be made to be in a vacuum state, or an adiabatic gas may be injected into the inner space to provide an insulation structure in the transparent panel assembly **60**.

(416) In more detail of the transparent panel assembly **60** with reference to the drawings, the transparent panel assembly **60** may have an outer appearance that is defined by the front panel **61** and the rear panel **65**, which define the front and rear surfaces of the transparent panel assembly **60**, and the outer spacer **67** connecting the front panel **61** to the rear panel **65**.

(417) Also, a display **62** and a light guide plate **64** may be disposed between the front panel **61** and the rear panel **65**. In addition, a first spacer **63** for supporting the display **62** and the light guide plate **64** may be further provided, and a display light **68** for irradiating light to the light guide plate **64** may be provided.

(418) The front panel **61** may have a size corresponding to that of the plate opening **511** and may have a size greater than that of the frame opening **701**. Thus, the periphery of the front panel **61** may be supported by the support frame **70**. Also, in a state in which the transparent panel assembly **60** is mounted, an end of the front panel **61** may come into contact with an end of the plate opening **511**, and a space may not be defined between the plate opening **511** and the front panel **61**.

(419) In detail, a front protrusion **613** that further protrudes outward than the rear panel **65** may be disposed on the front panel **61**. Due to structural characteristics of the front protrusion **613** inserted into and mounted on the front side of the outer plate **51**, the front protrusion **613** may further protrude from the rear panel **65** and the outer spacer **67** in upward/downward and left/right directions. Thus, the front panel **61** defining the front surface of the transparent panel assembly **60** may further extend to the outside of the frame opening **701** and thus may be stably supported by the support frame **70**. The rear panel **65** as well as the outer spacer **67** may be inserted into the frame opening **701**.

(420) Also, the support frame **70** and the outer spacer **67** of the transparent panel assembly **60** may be fastened and coupled to each other through a separate coupling structure or coupling members **78** such as a screw. Thus, when the transparent panel assembly **60** is mounted, the front protrusion **613** may be supported by the support frame **70**, and simultaneously, the support frame **70** may be coupled to the outer spacer **67** so that the heavy transparent panel assembly **60** is maintained in a stably fixed and mounted state even when the sub-door **50** is opened and closed.

(421) A bezel **611** may be disposed on a periphery of the rear surface of the front panel **61**. The bezel **611** may be formed by printing with an opaque color such as black so that the constituents such as the outer spacer **67**, the first spacer **63**, and the support frame **70** are not seen from the outside. The bezel **611** may have a predetermined width from an outer end of the front panel **61** to the first spacer **63**, which defines the see-through part **21** and is enough to cover the outer spacer **67**, the first spacer **63**, and the support frame **70**.

(422) A touch sensor **612** may be disposed on the rear surface of the front panel **61**. The touch sensor **612** may be formed on the rear surface of the front panel **61** in a printing manner and be configured to detect user's touch manipulation of the front panel **61**. Alternatively, the touch sensor **612** may be formed in various manners such as a film adhesion manner, rather than the printing manner, so that the user touches the front panel **61** to perform the touch input.

(423) A touch cable **601** connected to the touch sensor **612** may be disposed on the upper end of the front panel **61**. The touch cable **601** may be provided as a flexible film type cable such as a flexible flat cable (FFC) or a flexible print cable or flexible print circuit board (FPC). A printed circuit may be printed on the touch cable **601** to constitute at least a portion of a touch PCB **603**. Also, the touch cable **601** may be connected to the touch PCB **603** provided above the sub-door **50**.

(424) The touch cable **601** may be connected to the touch sensor **612** to extend upward. Also, the touch cable **601** may be configured so that a wire is disposed on a base made of a resin material such as a film and may extend upward along the rear surface of the front panel **61**. The touch cable **601** may be flexibly bent so that the touch cable **601** has a thin thickness and a wide width like a sheet.

(425) Also, the touch cable **601** may be provided as a film type. Thus, when the touch cable **601** is connected to the touch PCB **603**, an end of the touch cable **601** may be easily inserted into a connector of the touch PCB **603**. For this, the touch cable **601** may be bent several times, and the end of the touch cable **601** may be directed to the connector of the touch PCB **603**. Also, the touch cable **601** may be bent to be disposed along a well surface of an inner space of the sub-door **50** to provide an efficient arrangement in inner space of the sub-door **50**.

(426) Also, the display cable **605** and the display light cable **606** in addition to the touch cable **601** may have the same structure. As described above, the cables **601**, **605**, and **606**, each of which has a flat cable shape, may extend to an upper end of the transparent panel assembly **60**, and the cables **601**, **605**, and **606**, each of which has the thin thickness and the wide width, may be efficiently disposed on the sub-door **50**. In addition, a simple structure connected to the PCBs **601**, **605**, and **606** disposed in the upper portion of the sub-door **50** may be provided.

(427) The display **62** may be disposed on the rear surface of the front panel **61**. The display **62** may be provided as an LCD module for outputting a screen. Also, the display **62** may be transparent so that the user sees the inside through the display **62** when the screen is not outputted.

(428) A source board **621** may be disposed on one end of both left and right sides of the display **62**. The source board **621** may be configured to output a screen through the display **62** and connected to the display **62** and thus provided in an assembled state. Also, a portion of the source board **621** may also have a flexible film type cable structure.

(429) Also, the source board **621** may have a width less than a thickness of the transparent panel assembly **60** and be bent while the transparent panel assembly **60** is assembled. Here, the source board **621** may be disposed between the outer spacer **67** and the first spacer **63** and may come into contact with an inner surface of the outer spacer **67** while being perpendicular to the front panel **61**.

(430) Also, the source board **621** may be connected to a display cable **605**. The display cable **605** may be connected to a T-CON board **602** at an upper portion of the sub-door **50**.

(431) In detail, when the source board **621** is disposed on the rear surface of the display **62**, the source board **621** may be exposed to the outside through the see-through part **21** due to the characteristics of the display **62** that is transparent. Also, when the source board **621** has a structure that protrudes laterally, the sub-door **50** may increase in size.

(432) Thus, the source board **621** may be disposed on a peripheral end of the display **62** and may be provided between the outer spacer **67** and the first spacer **63**. Also, the source board **621** may have a size corresponding to that of the outer spacer **67** without out of a region of the outer spacer **67** in a state of being closely attached to the outer spacer **67**.

(433) The source board **621** may be constituted by two upper and lower boards **621** and respectively connected to the pair of display cables **605**. The display cable **605** may have a flexible and flat structure like the touch cable **601** and also have a structure that is freely bendable.

(434) The display cable **605** may extend along the peripheral surface of the transparent panel assembly **60** and pass through a sealant **608** defining the peripheral surface of the transparent panel assembly **60** to extend to the outside of the transparent panel assembly **60**.

(435) Also, the display cable **605** may be bent to extend along the peripheral surface of the transparent panel assembly **60**, i.e., be bent so that an end thereof extends upward from the transparent panel assembly **60**. Thus, the display cable **605** may be coupled to the T-CON board **602** at the upper side of the sub-door **50**.

(436) Both ends of the display **62** may be supported by the first spacer **63**. The first spacer **63** may have a rod shape extending from an upper end to a lower end of the display **62** and may be formed of aluminum.

(437) The light guide plate **64** may be disposed at the rear of the display and disposed to be spaced a predetermined distance from the display **62** by the first spacer **63**. Here, there may be a difference in depth feeling of the screen outputted from the display **62** according to the position of the light guide plate **64**.

(438) The light guide plate **64** may diffuse or scatter light emitted from the display light **68** and be made of various materials. For example, the light guide plate **64** may be made of a polymer material or formed by forming a pattern or attaching a film on a surface thereof. The light guide plate **64** may illuminate the display **62** from the rear side of the display **62** when the display light **68** is turned on. For this, the light guide plate **64** may have a plate shape having a size equal to or somewhat greater than that of the display **62**. The display light **68** may be disposed at a position

corresponding to each of upper and lower ends of the light guide plate **64**.

(439) Alternatively, when the display **62** is not provided, a separate glass or a heat insulating glass instead of the light guide plate **64** may be disposed.

(440) The rear panel **65** may be disposed at a rear side of the light guide plate **64**. The rear panel **65** may define the rear surface of the transparent panel assembly **60** and have a size greater than that of the light guide plate and less than that of the front panel **61**. Also, the rear panel **65** may have a size greater than that of the linear opening **561** to cover the linear opening **561**.

(441) A periphery of the rear panel **65** may further protrude outward from the outer spacer **67** to provide a rear panel protrusion **651**. The rear panel protrusion **651** may have a protruding portion which is seated on the door liner **56** when the transparent panel assembly **60** is mounted, and may define a space in which the sealant applied to the periphery of the sub-door **50** is filled.

(442) The rear panel **65** may be made of low- ϵ glass to realize thermal insulation. As a result, the rear panel **65** may prevent heat of cool air within the refrigerator from being transferred to the outside through the transparent panel assembly **60**.

(443) A pair of second spacers **66** may be disposed between the rear panel **65** and the light guide plate **64**. Each of the second spacers **66** may have a rectangular frame shape disposed along a periphery of the light guide plate **64** and adhere to the light guide plate **64** and the rear panel **65** to maintain a predetermined distance between the light guide plate **64** and the rear panel **65**. Also, a heat insulating glass **69** may be provided between the pair of second spacer **66**. A multilayered insulating layer may be provided between the light guide plate **64** and the rear panel **65** by the heat insulating glass **69**. Alternatively, a structure in which the light guide plate **64** and the rear panel **65** are fixed to each other by one second spacer **66** without the heat insulating glass **69** may be adopted as needed.

(444) Although the spacers **63**, **66**, and **67** have structures different from each other in this embodiment, the spacers **63**, **66**, and **67** may maintain a distance between the adjacent panels **61** and **65** and the light guide plate **64** and have various shapes such as a rod shape or a shape in which the moisture absorbent is accommodated into a shape.

(445) Also, the insulation panel **69** and the light guide plate **64** may be disposed between the front panel **61** and the rear panel **65**. Here, the insulation panel **69** and the light guide plate **64** may be plate-shaped members disposed between the front panel **61** and the rear panel **65** and may be lonely provided or may be provided together and also may be called intermediate panels. At least one or more intermediate panels may be provided. When a see-through part through which the inside is capable of being seen is provided, the intermediate panels may not be provided between the front panel **61** and the rear panel **65**.

(446) The distance between the front panel **61** and the light guide plate **64** may be maintained in fixed distance so as to output the screen of the display **62**. Also, the distance between the light guide plate **64** and the rear panel **65** may be determined according to a thickness of the sub-door **50** or the total thickness of the transparent panel assembly **60**. That is, the second spacer **66** may be adjusted in thickness to determine the total thickness of the transparent panel assembly **60** so as to be mounted to match a specification of the sub-door **50**.

(447) The rear panel **65** may come into contact with the door light **57**. Thus, a distance between the display **62** and the door light **57** may be determined according to the position of the rear panel **65**. A space behind the transparent panel assembly **60** may be illuminated by the door lights **57**, making it possible to visualize the storage space. Also, the door light **57** may serve as an auxiliary backlight of the display **62** in the turn-on state.

(448) A space between the light guide plate **64** and the rear panel **65** may be sealed by the second spacer **66**. Thus, a space between the second spacer **66** and the light guide plate **64** may become to a vacuum state, or an insulative gas such as argon may be injected for the thermal insulation to more improve the thermal insulation performance.

(449) In the state in which the rear panel **65** adheres to the second spacer **66**, an outer end of the

rear panel **65** may further extend outward from the second spacer **66**. Also, the outer spacer **67** may be mounted on the outer end of the rear panel **65** so that the rear panel **65** and the front panel **61** are fixed to each other.

(450) The outer spacer **67** may have a rectangular frame shape. The outer spacer **67** may connect the rear surface of the front panel **61** to the front surface of the rear panel **65** and also define the circumferential surface of the transparent panel assembly **60**.

(451) In detail, the outer spacer **67** may define a periphery of an outer portion of the transparent panel assembly **60** and also have a connection structure that is capable of allowing the front panel **61** to be maintained at a certain distance.

(452) The space between the front panel **61** and the rear panel **65**, i.e., the inner space of the outer spacer may be completely sealed by the coupling of the outer spacer **67**. Also, the inside of the outer spacer **67** may be more sealed by the sealant **608** applied to the periphery of the outer spacer **67**.

(453) The display **62** and the light guide plate **64** may be spaced apart from each other in a front and rear direction within the inside of the space that is sealed by the outer spacer **67**. The first and second spacers **63** and **66** for maintaining the distance of the light guide plate **64** may be also provided in the inner space of the outer spacer **67**.

(454) An additional insulation panel **69** may be further provided in the outer spacer **67**, or a multilayered glass structure may be provided in the outer spacer **67**. All of the above-described constituents may be provided in the space defined by the outer spacer **67**.

(455) That is, the overall outer appearance of the transparent panel assembly **60** may be defined by the front panel **61**, the rear panel **65**, and the outer spacer **67**, and all of the remaining constituents may be provided in the outer spacer **67**. Thus, the sealing may be performed only between the outer spacer **67**, the front panel **61**, and the rear panel **65** to completely seal the multilayered panel structure.

(456) Particularly, even through a plate-shaped structure such as the light guide plate **64** is further provided in the outer spacer **67**, when only the outer spacer **67** adheres to the front panel **61** and the rear panel **65**, the sealed structure of the transparent panel assembly **60** may be achieved. The sealed structure may maintain a minimal sealing point even in the multilayered structure due to the plurality of panel including the light guide plate **64**.

(457) Thus, introduction of external air into the transparent panel assembly or the dew condensation in the transparent panel assembly due to introduction of moisture may be minimized. Also, when the inside of the outer spacer **67** becomes in a vacuum state, or a gas for the thermal insulation is injected, the insulation layer may be provided in the whole multilayered structure within the transparent panel assembly **60** to more improve the thermal insulation performance.

(458) The transparent panel assembly **60** may be disposed in the sub door **50** so that the inside of the refrigerator is seen, and the screen is outputted, and also, the thermal insulation structure may be achieved in the multilayered panel structure at the minimum sealing point to secure the thermal insulation performance.

(459) Also, a space in which the display light **68** is mounted may be provided in an inner surface of the outer spacer **67**. The display light **68** may be mounted on each of the upper and lower ends of the outer spacer **67**. The light guide plate **64** may be disposed between the display lights **68** disposed on the upper and lower ends of the outer spacer **67**.

(460) Thus, light emitted through the display light **68** may be directed to an end of the light guide plate **64** and then travel along the light guide plate **64** so that the entire surface of the light guide plate **64** emits light.

(461) The display lights **68** disposed on the inner upper and lower ends of the transparent panel assembly **60** may be connected to a display light cable **606**. The display light cable **606** may have a flexible and flat shape like the touch cable **601** and the display cable **605**.

(462) The display light cable **606** may be connected to the display light **68** that is mounted inside

the outer spacer **67** to extend to the outside of the transparent panel assembly **60**.

(463) Also, the display light cable **606** may extend along the circumference of the transparent display **62** so that the display light cable **606** is not exposed through the transparent display **62**. Also, the display light cable **606** may extend upward in a state of being closely attached to the rear surface of the rear panel **65**. As occasion demands, the display light cable **606** may be bent in the state of adhering to the rear surface of the rear panel **65** and then may be connected to a docking PCB **604** disposed on the upper portion of the sub door **50**.

(464) Here, since the display light cable **606** extends in the state of being closely attached to a circumferential surface of the rear protrusion **651** of the rear panel **65**, when the sub door **50** is viewed from the outside, the display light cable **606** may not be exposed through the transparent panel assembly **60**.

(465) The sealant **608** may be applied to the circumference of the outer spacer **67**. The sealant **608** may be applied to form the circumferential surface of the transparent panel assembly **60**. That is, the sealant **691** may form a peripheral surface between the front panel **61** and the rear panel **65**.

(466) The sealant **608** may seal the transparent panel assembly **60** to prevent air from being introduced into the transparent panel assembly **60** and be made of a polysulfide (that is called a thiokol) material. As occasion demands, the sealant **691** may be made of a different sealant material such as silicon or urethane so that the sealant **691** comes into direct contact with the foaming solution that is injected to mold the insulation material **531**.

(467) The sealant **608** may maintain the coupling of the outer spacer **67**, the front panel **61**, and the rear panel **65** and completely seal the connected portions of the components to prevent water or moisture from being introduced. Also, the sealant **608** may be a portion, which comes into directly contact with the foaming solution when the insulation material **531** is molded, and protect the periphery of the transparent panel assembly **60**.

(468) Also, the sealant **608** may allow cables **601**, **605**, and **606** connected to the touch sensor **612**, the display panel **62**, and the display light **68** within the transparent panel assembly **60** to be accessible therethrough. The sealant **608** may cover outer surfaces of the cables **601**, **605**, and **606** to prevent water or moisture from being introduced through spaces through which the cables **601**, **605**, and **606** are accessible when the cables **601**, **605**, and **606** extend through the peripheral surface of the transparent panel assembly **60**.

(469) Also, a spacer protrusion **672** defining a space into which the sealant **608** is filled and a heater mounting part **673** on which a heater **532a** is mounted may protrude from the peripheral surface of the transparent panel assembly **60** coated with the sealant **608**, and the sealant **608** may be filled into a space defined between the spacer protrusion **672** and the heater mounting part **673**. A more detailed structure of the spacer protrusion **672** and the heater mounting part **673** will be described below again.

(470) FIG. **43** is a partial perspective view illustrating an arranged state of the display cable of the transparent panel assembly.

(471) As illustrated in the drawing, the display cable **605** may be connected to the source board **621** to extend upward. Then, the display cable **605** may extend along the periphery of the side surface of the transparent panel assembly **60** and then be connected to the T-CON board **602**.

(472) The display cable **605** may be connected to the source board **621** inside the transparent panel assembly **60**. The display cable **605** may be guided to the outside of the outer spacer **67** through the space between the rear panel **65** and the outer spacer **67**.

(473) In detail, a cable connection part **605a** is provided on the display cable **605**. The cable connection part **605a** may be introduced into the transparent panel assembly **60** through the space defined by the rear panel **65** and the end of the outer spacer **67** and then be connected to the source board **621** in the inner space of the transparent display **62**.

(474) The cable connectors **605a** may be guided to an outer surface of the transparent panel assembly **60** through a space between a gap of an adhesive member **671** allowing the rear panel **65**

and the outer spacer **67** to adhere to each other and the sealant **608**. Thus, the display cables **605** may pass through the sealed periphery of the sealed transparent panel assembly to be guided to the outside.

(475) The adhesive member **671** may also be provided between the front panel **61** and an end of the outer spacer **67**. The adhesive member **671** may have a thin thickness so that heat generated from the outer spacer **67** is sufficiently transferred to the front panel **61**. Alternatively, the outer spacer **67** may be coupled to the front panel **61** through a different method without adhering by the adhesive member **671**. Here, the outer spacer **67** may come into direct contact with the front panel to transfer heat.

(476) In this state, the display cables **605** may extend upwards in a bent state to come into contact with the outer surface of the transparent assembly **60** to which the sealant **608** is applied, and may be bent again to be connected to the T-CON board **602**. That is, the display cables **605** may extend to be connected to the T-CON board **602** while being exposed to the outside of the transparent panel assembly **60**.

(477) Also, the display cable **605** may be exposed to the outer surface of the transparent panel assembly **60** in the state of coating with the sealant **608**, and the spacer protrusion **672** and the heater mounting part **673** may be exposed between the sealants **608**. Thus, the transparent panel assembly **60** may be mounted on the door **50** in the state of being assembled. In the state in which the transparent panel assembly **60** is mounted on the door **50**, the process of fixing the transparent panel assembly **60** and for mounting the heater **532a** or connecting the mounted heater **532a** may be performed.

(478) FIG. **44** is a cross-sectional view illustrating a state in which the sealant is applied to both ends of the transparent panel assembly. Also, FIG. **45** is a cross-sectional view illustrating a state in which the sealant is applied to upper and lower ends of the transparent panel assembly. Also, FIG. **46** is a view illustrating a process of applying the sealant to the transparent panel assembly.

(479) As illustrated in the drawings, the sealant **608** may be applied to the periphery of both left and right surfaces and top and bottom surfaces of the transparent panel assembly **60**. The sealant **608** may be applied to a gap between the front panel **61** and the rear panel **65** and may be configured to cover the outer side of the outer spacer **67**.

(480) The transparent panel assembly **60** may be mounted in a state in which the sealant **608** is applied and may be supported by the support frame **70**. Thus, there is a limitation in that when the sealant **608** does not have a uniform surface, if the transparent panel assembly **60** is assembled, the transparent panel assembly **60** may be incorrectly assembled by interference with the support frame **70** or other adjacent components, or a failure may occur.

(481) In particular, when an interval between the front panel **61** and the rear panel **65** is large, it is not easy to uniformly apply the sealant **608**, and the sealant **608** may be biased to one side or may have an uneven surface in a local section.

(482) To prevent such a limitation, the spacer protrusion **672** and the heater mounting part **673** may be disposed on the outer surface of the outer spacer **67**. The spacer protrusion **672** and the heater mounting part **673** may be disposed in parallel to each other at positions spaced apart from each other to protrude at the same height. Also, the sealant **608** may be filled at the uniform height into the spaces between the front panel **61** and the rear panel **65** and between the spacer protrusion **672** and the heater mounting part **673**.

(483) The spacer protrusion **672** may be disposed on one side in a width direction of the outer spacer **67** and also be disposed at a position that is close to the front panel **61**. Here, the spacer protrusion **672** may be disposed between the heater mounting part **673** and the front panel **61**.

(484) Also, the spacer protrusion **672** may extend in a longitudinal direction of the outer spacer **67**. The spacer protrusion **672** may continuously extend from one end to the other end of the outer spacer **67** and may continuous along the periphery of the transparent panel assembly **60**.

Alternatively, if necessary, the spacer protrusions **672** having a predetermined length may be

disposed at a predetermined interval.

(485) After the sealant **608** is applied, an outer surface of the spacer protrusion **672** may be exposed to the peripheral surface of the transparent panel assembly **60**. Also, a plurality of coupling holes **672a** may be defined in the exposed outer surface of the spacer protrusion **672**. The plurality of coupling holes **672a** to which the coupling members **78** are coupled for the coupling of the transparent panel assembly **60** may be defined along the spacer protrusion **672**. It is preferable that the coupling holes **672a** are disposed along the spacer protrusion **672** and are located at a lower portion of the outer spacer **67**, which does not interfere with the cables **605**.

(486) Also, the heater mounting part **673** may be disposed on one side in a width direction of the outer spacer **67** and also be disposed at a position that is close to the rear panel **65**. That is, the heater mounting part **673** may be disposed between the rear panel **65** and the spacer protrusion **672**. Also, the heater mounting part **673** may extend in parallel to the spacer protrusion **672**, i.e., may continuously extend from one end to the other end of the outer spacer **67**. Also, the heater mounting part **673** may be provided in plurality, which are continuously disposed at a predetermined interval.

(487) Also, a heater groove **673a** may be defined in an outer surface of the heater mounting part **673**. The heater groove **673a** may be defined along the heater mounting part **673** and have a size and shape corresponding to be inserted into and mounted on the outer portion of the heater **532a**. The heater groove **673a** may have a size so that the heater **532a** is press-fitted and fixed thereto or is fixed by a separate fixing member.

(488) Also, the heater groove **673a** may be exposed to the outside so that the heater **532a** is mounted in the exposed heater groove **673a** in the state in which the transparent panel assembly **60** is mounted on the door **50**. That is, the heater mounting part **673** may be disposed closer to the rear panel **65** than the front panel **61** so that the heater mounting part **673** is exposed to the outside when the transparent panel assembly **60** is mounted on the door **50**. Thus, when the heater **532a** is mounted, the heater **532a** may not interfere with other constituents within the door **50** to improve convenience in work. Alternatively, the transparent panel assembly **60** may be mounted on the door **50** in the state in which the heater **532a** is mounted in the heater groove **673a**.

(489) The outer spacer **67** may be made of a metal material, particularly, made of an aluminum material having superior heat transfer performance. Thus, when the heater **532a** generates heat in the state in which the heater **532a** is mounted on the heater mounting part **673**, the outer spacer **67** may also generate heat to transfer the generated heat from the outer spacer **67** to the front panel **61**.

(490) That is, heat may be generated from an end of the outer spacer **67** coming into contact with the front panel **61**. Thus, when compared with a structure in which the heater **532a** itself comes into contact with the front panel **61**, a wider area of the end of the outer spacer **67** may come into contact with the front panel **61** to provide a more amount of heat to the front panel **61**, thereby effectively preventing dew condensation from being generated.

(491) In addition, an area on which the outer spacer **67** comes into contact with the front panel **61** may be an area that is substantially close to the outside of the see-through part **21** and also the innermost area to be heated while preventing the heater **532a** from being exposed. That is, it is preferable that the end of the outer spacer **67** comes into contact with the area of the bezel **611** of the front panel **61**. Thus, the outer spacer **67** may not be exposed to the outside by being covered by the bezel **611**.

(492) Also, the position at which the outer spacer **67** is disposed may be substantially a non-insulation region. In detail, the insulation space of the transparent panel assembly **60** is defined inside the outer spacer **67**, and the periphery of the door **50** is thermally insulated by the insulator **531**. On the other hand, a constituent for the insulation is not provided from the outer spacer **67** to the position of the insulator **531**, and thus, the dew condensation may be generated on the front surface of the transparent panel assembly **60** adjacent to the non-insulation region R.

(493) In addition, the non-insulation region R may be an area in which the distance between the door liner **56** and the front panel **61** is close to cause insufficient thermal insulation. Thus, there is a

high possibility that dew condensation occurs on the front surface of the front panel **61** at the corresponding position.

(494) The outer spacer **67** may be disposed around the transparent panel assembly **60** which is likely to cause the dew condensation due to the non-insulation region R. and the heater **532a** may be mounted on the outer spacer **67** to heat the non-insulation region R by heat generated by the outer spacer itself, thereby preventing the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(495) The heater **532a** may have a wire shape, and a generally-used sheath heater may be used as the heater **532a**. The heater **532a** may have a diameter that is enough to be inserted into the heater groove **673a** and be disposed over entire four surfaces of the periphery of the transparent panel assembly **60**.

(496) The heater **532a** may be disposed on only both the left and right surfaces and the bottom surface of the periphery of the transparent panel assembly **60**. That is, since the upper portion of the transparent panel assembly **60** is heated by heat generated when the plurality of PCBs **602**, **603**, and **604** operate, the heater **532a** may not be provided on at least a portion of the top surface of the periphery of the transparent panel assembly **60**.

(497) Also, the spacer protrusion **672** and the heater mounting part **673** may protrude up to a height corresponding to that of the rear panel **65**. Thus, the space between the front panel **61** and the rear panel **65** may be divided into four spaces by the spacer protrusion **672** and the heater mounting part **673**. The sealant **608** may be filled into each of the spaces to the same height.

(498) As illustrated in FIG. **14**, to allow the sealant **608** to have a uniform height, after the sealant **608** is filled in the space **673** defined by the spacer protrusion **672** and the heater mounting part **673**, a level of the sealant **608** may be adjusted using a separate jig or a scraper S.

(499) In detail, when the jig or the scraper S comes into contact with the peripheral surface of the transparent panel assembly **60** in a state in which the sealant **608** is filled between the front panel **61** and the rear panel **65**, a lower end of the jig or the scraper S may come into contact with the rear panel **65**, the spacer protrusion **672**, and a protruding end of the heater mounting part **673**, which have the same height. Also, the other side of the jig or the scraper S may come into contact with the rear surface of the front panel **61**. In this state, when the jig or the scraper S moves, the sealant **608** may be filled in each of the spaces between the rear panel **65** and the heater mounting part **673** and between the spacer protrusion **672** and the front panel **61** by the height of the spacer protrusion **672**, the heater mounting part **673**, and the rear panel **65**, and the remaining portion may be removed by the jig or the scraper S.

(500) Thus, when the jig or the scraper S moves along the periphery of the transparent panel assembly **60**, the sealant **608** may be applied to the periphery of the transparent panel assembly **60** at a uniform height. Also, when the transparent panel assembly **60** is mounted, the sealant **608** may not interfere with the support frame **70** or other constituents.

(501) As illustrated in FIGS. **12** and **13**, the spacer protrusion **672** may be disposed at the periphery on both the left and right surfaces and the upper and lower surface of the transparent panel assembly **60**. Thus, the sealant **608** may be applied to the entire periphery of the transparent panel assembly **60**, and all upper, lower, left, and right portions of the periphery of the transparent panel assembly **60** may be stably fixed to the support frame **70**.

(502) Also, as illustrated in FIG. **14**, in the state in which the sealant **608** is applied to the uniform height, the end of the heater mounting part **673**, i.e., the heater groove **673a** and the end of the spacer protrusion **672** may be exposed. Thus, the transparent panel assembly and the support frame **70** may be coupled to each other through the mounting of the heater **532a** and the coupling of the coupling member **78**.

(503) FIG. **47** is a cutaway perspective view illustrating a state in which the outer plate and the support frame are coupled to each other according to the tenth embodiment of the present disclosure. Also, FIG. **48** is an exploded cutaway perspective view illustrating a coupled structure

between the outer plate and the support frame.

(504) In more detail of the coupling structure between the support frame **70** and the outer plate **51** with reference to the drawings, the bent plate part **514** may be bent along the plate opening **511** defined at a center of the outer plate **51**.

(505) The support frame **70** may be mounted on the rear surface of the outer plate **51**. The support frame **70** may be disposed along the periphery of the plate opening **511**.

(506) The side frame **73** may be disposed on both left and right ends of the plate opening **511**. Here, the bent plate part **514** may be inserted into the plate accommodation groove **75**.

(507) A guide rib **751** including the vertical part **527a** and the inclined part **527b** may be disposed inside the plate accommodating groove **75**. Thus, while the bent plate part **514** is inserted into the plate accommodating groove **75**, the bent plate part **514** may be inserted while moving along the inclined part **527b**, and the inner surface of the bent plate part **514** may be supported by the vertical part **527a**.

(508) The bent plate part **514** may be guided toward the inside of the plate opening **511** by the guide rib **751** and may maintain a position thereof in a state in which the bent plate part **514** is completely inserted into the plate accommodating groove **75**. Here, the guide rib **751** may support the bent plate part **514** in a manner to slightly press the bent plate part **514** from the inner side and may prevent separation or moving of the outer plate **51**.

(509) Thus, as illustrated in FIG. **21**, in the state in which the transparent panel assembly **60** is mounted, the bent plate part **514** is disposed inside the plate accommodating groove **75** and may be maintained to come into close contact with the outer end of the front panel **61**. Due to such a structure, an interval or gap between the transparent panel assembly **60** and the outer plate **51** on the front surface of the sub-door **50** may not be virtually seen, and a boundary of the transparent panel assembly **60** and a boundary of the outer plate **51** may completely come into close contact with each other so that the entire outer appearance of the front surface of the sub-door **50** has a sense of unity.

(510) Also, a guide insertion part **514a** may be disposed on one side of the bent plate part **514** so that the outer plate **51** is mounted on the support frame **70** at an accurate position. The guide insertion part **514a** may be disposed on an end of the bent plate part **514** so as to have a predetermined width and may pass through the support frame **70**.

(511) Also, an insertion guide hole **753** through which the guide insertion part **514a** passes may be defined in the support frame **70**. The insertion guide hole **753** may be defined in the bottom surface of the plate accommodating groove **75** and may have a size allowing the guide insertion part **514a** to pass therethrough.

(512) Thus, when the outer plate **51** and the support frame **70** are coupled to each other, the outer plate **51** and the support frame **70** may be aligned with each other so that the guide insertion part **514a** passes through the insertion guide hole **753**, and the bent plate part **514** is disposed inside the plate accommodating groove **75** at an accurate position.

(513) When the bent plate part **514** is inserted into the plate accommodating groove **75** at the accurate position, the restraint boss **752** disposed inside the plate accommodating groove **75** may be coupled to the restrainer **514b** disposed in the bent plate part **514**. In a state in which the bent plate part **514** is completely inserted into and fixed to the plate accommodating groove **75**, the restraint bosses **752** and the restrainers **514b** may be coupled to each other so that the bent plate part **514** is maintained in an inserted state.

(514) The plurality of guide insertion parts **514a** and the plurality of restrainers **514b** may be disposed in the bent plate part **514** at predetermined intervals. Also, the plurality of guide insertion parts **514a** and the plurality of restrainers **514b** may be disposed throughout the bent plate part **514**.

(515) In this state, an adhesive or an adhesive member are applied to the plate support **74** so that a state in which the plate support **74** is fixed and mounted to the rear surface of the outer plate **51** is maintained. Thus, even when a foam liquid is injected into the sub-door **50**, a position at which the

support frame **70** is fixed and mounted onto the outer plate **51** may be maintained.

(516) In the state in which the support frame **70** is mounted on the outer plate **51**, the transparent panel assembly **60** may be inserted and mounted from the front side to the rear side of the plate opening **511**. Here, in a state in which the rear panel **65** having a narrow width is firstly inserted and the transparent panel assembly **60** is inserted, the rear surface of the front panel **161** may be seated on the panel support **76**.

(517) Also, in a state in which the transparent panel assembly **60** is completely inserted and mounted, the coupling member **78** coupled while passing through the blocking part **77** may be coupled to the coupling hole **672a** of the outer spacer **67**. The periphery of the transparent panel assembly **60** may be coupled to the frame coupling part **77** by the plurality of coupling members **78**, and the transparent panel assembly **60** may be fixed and mounted.

(518) Thus, the transparent panel assembly **60** may be firmly mounted even in a state in which an adhesive structure of the periphery of the front panel **61** and the panel support **76** is not provided and may be maintained in a stable mounted state even when an impact is applied thereto while the sub-door **50** is opened and closed.

(519) Due to such a structure, when a limitation occurs in the transparent panel assembly **60**, and thus a follow-up service is required, the transparent panel assembly **60** may be easily disassembled. Also, when the transparent panel assembly **60** is disassembled, an adhesive or an adhesive member is not applied to the bezel **611** on the periphery of the front panel **61**. Thus, the transparent panel assembly **60** is easily separated, and the bezel **611** is prevented from being damaged by the adhesive or the adhesive member as well. Thus, the follow-up service may be easily performed, and the not-damaged transparent panel assembly **60** having a high price may be reused after the follow-up service.

(520) FIG. **49** is a cutaway perspective view taken along line **49-49'** of FIG. **38**. Also, FIG. **50** is a cross-sectional view taken along line **50-50'** of FIG. **38**.

(521) As illustrated in the drawings, in a state in which the outer plate **51** and the transparent panel assembly **60** are mounted on the support frame **70**, the transparent panel assembly **60** may be fixed and mounted onto the support frame **70** through the coupling member **78**. Also, the door liner **56** is coupled, and the door lights **57** and the gasket **503** are mounted so that the sub-door **50** is assembled.

(522) Also, in the state in which the transparent panel assembly **60** is mounted, the end of the heater mounting part **673** may be exposed to the outside, and the heater **523a** may be mounted in the heater groove **673a**. In the state in which the transparent panel assembly **60** is mounted, the heater **532a** may be mounted around the transparent panel assembly **60**, and an electric wire for supplying power may be connected.

(523) Also, a shielding member **79** may be attached to the frame coupling part **77** and the door liner **56** or the rear panel **65**. The shielding member **79** may be made of an attachable material such as a tape to completely block a gap between the shielding member **79** and the door liner **56** or the rear panel **65**. Thus, the foam liquid filled inside the sub-door **50** may be completely prevented from being introduced toward the transparent panel assembly **60**.

(524) Due to the prevention of the introduction of the foam liquid by the shielding member **79**, the foam liquid may be prevented from being polluted or being stained with the display cables **605** guided along the peripheral surface of the transparent panel assembly **60**, that is, an outer surface of the sealant **608**. That is, the display cables **605** may be disposed between the blocking part **77** and the sealant **608**. Thus, even when the foam liquid is injected into the sub-door **50**, the foam liquid is prevented from being introduced toward the display cables **605** by the blocking part **77**. Also, the foam liquid is not stained with the display cables **605**, and thus, even when the transparent panel assembly **60** is replaced or is separated for the follow-up service, the display cables **605** may be reused without being damaged.

(525) A foam solution may be injected into the assembled sub-door **50** to form the insulator **531**.

The insulator **531** may be filled in the outer side of the transparent panel assembly **60** to insulate a peripheral space of the sub-door **50**. Also, an insulation panel **69** may be provided, or a sealed insulation layer may be disposed between the front panel **61** and the rear panel **65** of the transparent panel assembly **60** so that the front panel **61** and the rear panel **65** may be insulated from each other. Thus, the insulation may be achieved throughout the entire surface of the sub-door **50**. Alternatively, the insulator **531** may be previously molded and then inserted into and mounted on the periphery of the sub-door **50** after the transparent panel assembly **60** is mounted.

(526) Hereinafter, an operation of the transparent panel assembly will be described in more detail with reference to the accompanying drawings.

(527) FIG. **51** is a transverse cross-sectional view of the main door and the sub-door. Also, FIG. **52** is an enlarged view illustrating a portion C of FIG. **51**. Also, FIG. **53** is an enlarged view illustrating a portion D of FIG. **51**. Also, FIG. **54** is a longitudinal cross-sectional view of the main door and the sub-door. Also, FIG. **55** is an enlarged view illustrating a portion E of FIG. **54**. Also, FIG. **56** is an enlarged view illustrating a portion F of FIG. **54**.

(528) As illustrated in the drawings, in a state in which the locking member **593** of the opening device **59** is inserted into a latch hole **421**, the sub-door **50** may be maintained in a closed state. In this state, the door light **57** may be maintained in a turn-off state. An opened or closed state of the sub-door **50** may be detected through a door switch that is separately provided.

(529) In the turn-off state of the door light **57**, as illustrated in FIG. **1**, the rear space of the sub door **50** may be dark, and thus, the inside of the refrigerator **1** may not be seen through the see-through part **21**. Thus, in the closed state of the sub-door **50**, if separate manipulation is not performed, the door light **57** may be maintained in the turn-off state, and the inside of the refrigerator **1** may not be seen through the see-through part **21**.

(530) In this state, the user may touch-manipulate the front panel **51** to turn on the door light **57**. When the door light **57** is turned on, light emitted from a lighting module **575** may be irradiated to positions of both rear left and right sides of the rear panel **65**, which face each other.

(531) The door light **57** may extend from the upper end to the lower end of the rear panel **65**. That is, the light emitted by the door light **57** may illuminate the entire rear region of the rear panel **65** from both the left and right sides of the rear panel **65**.

(532) Here, when the display light **68** is in the turn-on state together with the door light **57**, light may be emitted upward and downward by the display light **68**, and thus the light may be irradiated from left and right sides by the door light **57**. As a result, the light may be emitted to the see-through part **21** in all directions to maximally brighten up an area of the see-through part **21**.

(533) The door light **57** may emit light in directions facing each other in a state of being close to the rear panel **65**. The light emitted by the door light **57** may brighten up an inner case of the accommodation case **43** and also brighten up the front region over the rear panel **65**. Thus, as illustrated in FIG. **28**, the door light **57** may serve as a lighting for brightening up the inner space of the refrigerator **1**, which is seen through the see-through part **21** and also serve as an auxiliary backlight for allow the display **62** to be more clearly displayed.

(534) That is, in a state in which a screen is being outputted through the display **62**, the inner space of the refrigerator **1**, i.e., the rear space of the sub door **50** may be selectively seen through the see-through part **21**. To allow the rear space of the sub door **50** to be seen through the see-through part **21**, the door light **57** may be turned on.

(535) A turn on/off combination of the display light **68** and the door light **57** may be variously realized according to a degree of seeing of the inside of the accommodation case **43** through the see-through part **21**.

(536) Also, when the user manipulates the front panel **61** disposed on the front surface of the refrigerator **1**, the display light **68** may be turned on to turn on the display **62**. Thus, the transparent panel assembly **60** may output a screen. Here, the manipulation of the front panel **61** may be inputted as one of a specific position, the touch number, or a pattern. As occasion demands, a

separate physical button or sensor may be used to detect the user's manipulation.

(537) A screen for displaying a state of the refrigerator **1** and manipulating may be outputted on the display **62**. Here, various screens for information with respect to accommodated foods may be outputted by using Internet, image output external input devices, or the like.

(538) In detail, the display light **69** disposed on each of the upper and lower ends of the light guide plate **64** may be turned on together with the display **62** by the user's manipulation. The light guide plate **64** may irregularly reflect and diffuse light of the display light **68** by the turn-on of the display light **68** to emit light having generally uniform brightness to the front display **62**.

(539) Also, light may be emitted to the display **62** from the rear side of the display **62** by the light guide plate **64**, and simultaneously, a screen based on inputted image information may be outputted on the display **62**. Thus, the user may confirm the clearly outputted screen through the see-through part **21**.

(540) The operation of the display **62** and the operations of the door lights **57** may be controlled by the PCBs **602**, **603**, and **604** such as the T-CON board **602** or the docking PCB **604** above the sub-door **50**. Also, these PCBs **602**, **603**, and **604** may be arranged on the rear space of the sub-door **50**, which is partitioned by the barrier **711** defining the upper end of the support frame **70**. Also, the insulator **531a** may be filled in a front space of the sub-door **50**, which is partitioned by the barrier **711**, and thus dew condensation may be prevented from being generated on an upper side of the front surface of the sub-door **50**.

(541) Also, when the inside of the refrigerator is cooled and maintained at a set temperature by the operation of the refrigerator **1**, dew condensation may be generated on the front surface of the transparent panel assembly **60**, which corresponds to the non-insulation region in which the insulation is weak.

(542) The heater **532a** may operate to prevent the dew concentration from being generated on the transparent panel assembly **60**. The heater **532a** may be in the turn-on state and also repeatedly turned on/off for a set time.

(543) When the heater **532a** is turned on to generate heat, the outer spacer **67** on which the heater **532a** is mounted may be heated. The outer spacer **67** may be made of a metal material to transfer heat of the heater **532a** along the outer spacer **67**. Thus, the periphery of the front panel **61** coming into contact with the outer spacer **67** may be heated.

(544) Here, when compared with the heater **532a**, the end of the outer spacer **67** coming into contact with the front panel **61** may increase in surface area. Thus, the relatively wide area of the front panel **61** may be heated.

(545) As illustrated in FIGS. **26**, **27**, **29**, and **30**, the heater **532a** may be disposed on all both top/bottom and left/right surfaces of the outer spacer **67** and also disposed along the peripheral surface of the transparent panel assembly **60**.

(546) Thus, an edge of the front surface of the front panel **61** coming into contact with the outer spacer **67** may be heated on the whole, and an edge of the front panel **61** corresponding to the non-insulation region **R** may be heated to prevent the dew condensation from being generated. Also, one side of the front panel **61** coming into contact with the outer spacer **67** may be disposed inside the bezel **611** to prevent the heater **532a** and the outer spacer **67** from being exposed to the outside.

(547) In addition to the foregoing embodiment, a refrigerator according to various embodiments may be exemplified.

(548) In the eleventh embodiment of the present disclosure, the heater mounting part on which the heater is mounted is disposed on the end of the outer frame coming into contact with the front panel.

(549) An eleventh embodiment is the same as the abovementioned embodiments except for constituents of the heater mounting part, and thus, the same constituent as those according to the foregoing embodiments may be denoted by the same reference numeral, and its detailed description will be omitted. In addition, not-illustrated reference numerals will be the same as those of the

constituents illustrated in the drawings in the above-described embodiments.

(550) FIG. 57 is a cross-sectional view of a door according to an eleventh embodiment of the present disclosure.

(551) Referring to the drawing, an outer peripheral shape of the door 50 may be defined by the bent outer plate 51 made of a metal material. The outer plate 51 may define the front surface and a periphery of the side surfaces of the door 50. Also, the door liner 56 defining the rear surface of the door 50 is coupled to the outer plate 51, and the transparent panel assembly 60 is provided in openings of the outer plate 51 and the door liner 56 so that the interior of the refrigerator 1 is selectively seen. Also, the insulator 531 may be filled into the periphery of the door 50 outside the transparent panel assembly 60.

(552) The transparent panel assembly 60 may include the front panel 61 defining the front surface thereof, the rear panel 65 defining the rear surface thereof, and the insulation panel 69 between the front panel 61 and the rear panel 65. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel 61, and thus the interior of the refrigerator 1 may be selectively visualized according to whether the door lights 57 or a lamp in the refrigerator 1 is turned on or off.

(553) Also, the display 62 may be provided on the rear surface of the front panel 61. Here, the light guide plate 64 may be provided at the rear of the display 62. Also, the insulation panel 69 may be omitted if necessary or provided in plurality.

(554) A pair of second spacers 66 may be provided between the rear panel 65, the insulation panel 69, and the light guide plate 64 to support the rear panel 65, the insulation panel 69, and the light guide plate 64, and a first spacer 63 may be disposed between the light guide plate 64 and the front panel 61 to support the light guide plate 64 and the front panel 61. Also, an outer spacer 67 may be disposed outside the first spacer 63 and the second spacers 66. The outer spacer 67 may be configured to support the front panel 61 and the rear panel 65. At least one of a space between the first spacer 63 and the second spacer 66 or a space between the outer spacers 67 may be in a vacuum state, or an adiabatic gas may be injected into the space to form an insulation space.

(555) The spacer protrusion 672 may be disposed outside the outer spacer 67. Also, the spacer protrusion 672 may be coupled to the support frame 70 by the coupling member 78.

(556) Also, a sealant for sealing the peripheral surface of the transparent panel assembly 60 may be applied into the space between the front panel 61 and the rear panel 65 with respect to the spacer protrusion 672.

(557) A heater mounting part 675 may be disposed on the front end of the outer spacer 67, i.e., an end of the outer spacer 67 coming into contact with the front panel 61. Also, a heater groove 675a into which the heater 532a is inserted may be defined in the heater mounting part 675. The heater 532a mounted in the heater groove 675a may be disposed in a direction contacting or facing the front panel 61.

(558) An adhesive member 671 may be further disposed on the front end of the outer spacer 67. The front end of the outer spacer 67 may adhere to be fixed to the front panel 61. Alternatively, the adhesive member 671 may not be provided on the end of the outer spacer 67. Thus, the heater groove 675a and the end of the outer spacer 67 may come into direct contact with the rear surface of the front panel 61.

(559) When the heater 532a generates heat, the heat of the heater 532a may be transferred to the outer spacer 67 made of a metal material. The outer spacer 67 may come into contact with the front panel 61 on a wider area than that of the heater 532a. Thus, the heat transferred to the outer spacer 67 may heat a wider area when compared with a structure in which the front panel 61 is heated by only the heater 532a. In addition, the heating area may increase due to the heating of the outer spacer 67 in addition to the direct heating of the heater 532a to more effectively heat the front panel 61.

(560) Thus, the non-insulation region between the insulation space of the transparent panel

assembly **60** and the insulators of the door **50** may be effectively heated to effectively prevent the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(561) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51**, a panel support part **76** supporting the periphery of the front panel **61**, and a frame coupling part **77** coupled to the outer spacer **67**.

(562) The plate support part **74** may adhere to the rear surface of the outer plate **51** by the adhesive member. Here, an end of the plate support part **74** may be disposed to correspond to the end of the outer plate **51**, in which the opening is defined.

(563) Also, the panel support part **76** may be stepped on the plate support part **74** to allow the periphery of the front panel **61** that further protrudes outward to be seated. Here, the panel support part **76** may be stepped by a thickness of the plate support part **74** and the front panel **61**.

(564) Thus, in the state in which the transparent panel assembly **60** is mounted, the front surfaces of the outer plate **51** and the front panel **61** may have the same height and be disposed on the same plane. Also, the end of the outer plate **51** and an outer end of the front panel **61** may come into contact with each other. Thus, when viewed from the outside, a gap between the outer plate **51** and the front panel **61** may not be seen.

(565) The frame coupling part may extend from the panel support part **76**, i.e., may vertically extend from the end of the panel support part **76** to a position passing through the spacer protrusion **672**. Thus, the coupling member **78** passing through the frame coupling part **77** may be coupled to the spacer protrusion **672** to more firmly fix the transparent panel assembly **60** to the support frame **70**. Thus, the transparent panel assembly **60** may adhere to the support frame **70** without a separate adhesive.

(566) That is, the transparent panel assembly **60** may be fixed and mounted onto the support frame **170** by the blocking part **173**. Thus, the transparent panel assembly **60** may be firmly fixed, the transparent panel assembly **60** may be separated, and serviceability may be improved.

(567) In addition to the foregoing embodiment, a refrigerator according to various embodiments may be exemplified.

(568) In the twelfth embodiment of the present disclosure, the heater mounting part is disposed on the front portion of the outer spacer, and the spacer protrusion is disposed on the rear portion.

(569) The twelfth embodiment is the same as the abovementioned embodiments except for constituents of portions of the outer spacer and the support frame, and thus, the same constituent as those according to the foregoing embodiments may be denoted by the same reference numeral, and its detailed description will be omitted.

(570) FIG. **58** is a cross-sectional view of a door according to a twelfth embodiment of the present disclosure.

(571) Referring to the drawing, the door **50** may be provided by coupling the outer plate and the door liner. The transparent panel assembly **60** may be provided in the openings of the outer plate **51** and the door liner **56** so that the interior of the refrigerator **1** is selectively seen. Also, the insulator **531** may be filled into the periphery of the door **50** outside the transparent panel assembly **60**.

(572) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(573) Also, the display **62** may be provided on the rear surface of the front plate. Here, the light guide plate **64** may be provided at the rear of the display **62**. Also, the insulation panel **69** may be omitted if necessary or provided in plurality.

(574) A pair of second spacers **66** may be provided between the rear panel **65**, the insulation panel **69**, and the light guide plate **64** to support the rear panel **65**, the insulation panel **69**, and the light

guide plate **64**, and a first spacer **63** may be disposed between the light guide plate **64** and the front panel **61** to support the light guide plate **64** and the front panel **61**. Also, an outer spacer **67** may be disposed outside the first spacer **63** and the second spacers **66**. The outer spacer **67** may be configured to support the front panel **61** and the rear panel **65**. At least one of a space between the first spacer **63** and the second spacer **66** or a space between the outer spacers **67** may be in a vacuum state, or an adiabatic gas may be injected into the space to form an insulation space.

(575) The heater mounting part **673** and the spacer protrusion **672** may be disposed outside the outer spacer **67**. The heater mounting part **673** and the spacer protrusion **672** may be spaced apart from each other in the front and rear directions and protrude outward at the same height. Also, the sealant **608** may be applied between the front panel **61** and the rear panel **65** and between the spacer protrusion **672** and the heater mounting part **673**.

(576) The heater mounting part **673** may be disposed at a position that is close to the front panel **61** to protrude between the spacer protrusion **672** and the front panel **61**. Also, the heater **532a** may be inserted into the heater groove **673a** defined in the heater mounting part **673**.

(577) Also, the spacer protrusion **672** may be disposed further rearward than the heater mounting part **673** to protrude between the heater mounting part **673** and the rear panel **65**. Also, the spacer protrusion **672** may be coupled to the support frame **70** by the coupling member **78**.

(578) Due to the above-described structure, before the transparent panel assembly **60** is fixed and mounted on the door **50**, the heater **532a** is mounted on the heater mounting part **673**. Also, the transparent panel assembly **60**, in which the heater **532a** is mounted, may be coupled to the support frame **70** by the coupling member **78** and then fixed and mounted on the door **50**.

(579) When the heater **532a** generates heat, the heat of the heater **532a** may be transferred to the outer spacer **67** made of a metal material. The outer spacer **67** may come into contact with the front panel **61** on a wider area than that of the heater **532a**. Thus, the heat transferred to the outer spacer **67** may heat a wider area when compared with a structure in which the front panel **61** is heated by only the heater **532a**.

(580) Thus, the non-insulation region R between the insulation space of the transparent panel assembly **60** and the insulators **531** of the door may be effectively heated to effectively prevent the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(581) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51**, a panel support part **76** supporting the periphery of the front panel **61**, and a frame coupling part **77** coupled to the outer spacer **67**.

(582) The frame coupling part **77** may vertically extend from the panel support part **76** and also may extend further rearward than at least the spacer protrusion **672**. Thus, the coupling member **78** passing through the frame coupling part **77** may be coupled to the spacer protrusion **672**.

(583) In addition to the foregoing embodiment, a refrigerator according to various embodiments may be exemplified.

(584) In the thirteenth embodiment, a protrusion protrudes from the outer spacer, and also, the heater is mounted on the protrusion, and support frame is coupled to the protrusion.

(585) The thirteenth embodiment is the same as the abovementioned embodiments except for constituents of the outer spacer, and thus, the same constituent as those according to the foregoing embodiments may be denoted by the same reference numeral, and its detailed description will be omitted.

(586) FIG. **59** is a cross-sectional view of a door according to a thirteenth embodiment of the present disclosure.

(587) Referring to the drawing, the door **50** may be provided by coupling the outer plate and the door liner. The transparent panel assembly **60** may be provided in the openings of the outer plate **51** and the door liner **56** so that the interior of the refrigerator **1** is selectively seen. Also, the insulator may be filled into the periphery of the door **50** outside the transparent panel assembly **60**.

(588) The transparent panel assembly **60** may include the front panel **61** defining the front surface

thereof, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(589) Also, the display **62** may be provided on the rear surface of the front panel **61**. Here, the light guide plate **64** may be provided at the rear of the display **62**. Also, the insulation panel **69** may be omitted if necessary or provided in plurality.

(590) A pair of second spacers **66** may be provided between the rear panel **65**, the insulation panel **69**, and the light guide plate **64** to support the rear panel **65**, the insulation panel **69**, and the light guide plate **64**, and a first spacer **63** may be disposed between the light guide plate **64** and the front panel **61** to support the light guide plate **64** and the front panel **61**. Also, an outer spacer **67** may be disposed outside the first spacer **63** and the second spacers **66**. The outer spacer **67** may be configured to support the front panel **61** and the rear panel **65**. At least one of a space between the first spacer **63** and the second spacer **66** or a space between the outer spacers **67** may be in a vacuum state, or an adiabatic gas may be injected into the space to form an insulation space.

(591) The protrusion **676** may be disposed outside the outer spacer **67**. The protrusion **676** may protrude outward between the front panel **61** and the rear panel **65**. Also, the protrusion **676** may have a height corresponding to the rear panel, and the sealant **608** may be filled between the protrusion **676**, the front panel **61**, and the rear panel **65**.

(592) Also, the protrusion **676** may be disposed at a position corresponding to the frame coupling part **77** of the support frame **70** and have a coupling hole **676a** to which the coupling member passing through the frame coupling part **77** is coupled.

(593) Also, a heater groove **676b** into which the heater **532a** is inserted may be defined in one side of a coupling hole **676a**. That is, the coupling hole **676a** and the heater groove **676b** may be defined in an outer end of the protrusion **676** so that the support frame **70** is coupled, and the heater **532a** is mounted through the protrusion **676**.

(594) The heater groove **676b** may be disposed further rearward than the coupling hole **676a** and also be disposed further rearward than the frame coupling part **77** so as to be exposed to the outside in the state in which the coupling member **78** is coupled. That is, in the state in which the transparent panel assembly **60** is fixed and mounted on the door **50**, the heater **532a** may be mounted in the heater groove **676b**.

(595) When the heater **532a** generates heat, the heat of the heater **532a** may be transferred to the outer spacer **67** made of a metal material. The outer spacer **67** may come into contact with the front panel **61** on a wider area than that of the heater **532a**. Thus, the heat transferred to the outer spacer **67** may heat a wider area when compared with a structure in which the front panel **61** is heated by only the heater **532a**.

(596) Thus, the non-insulation region R between the insulation space of the transparent panel assembly **60** and the insulators **531** of the door **50** may be effectively heated to effectively prevent the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(597) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51**, a panel support part **76** supporting the periphery of the front panel **61**, and a frame coupling part **77** coupled to the outer spacer **67**.

(598) In addition to the foregoing embodiment, a refrigerator according to various embodiments may be exemplified.

(599) In the fourteenth embodiment of the present disclosure, only the heater mounting part is provided on the outer spacer.

(600) The fourteenth embodiment is the same as the abovementioned embodiments except for constituents of the outer spacer, and thus, the same constituent as those according to the foregoing

embodiments may be denoted by the same reference numeral, and its detailed description will be omitted.

(601) FIG. **60** is a cross-sectional view of a door according to a fourteenth embodiment of the present disclosure.

(602) Referring to the drawing, the door **50** may be provided by coupling the outer plate and the door liner. The transparent panel assembly **60** may be provided in the openings of the outer plate **51** and the door liner **56** so that the interior of the refrigerator **1** is selectively seen. Also, the insulator **531** may be filled into the periphery of the door **50** outside the transparent panel assembly **60**.

(603) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof, the rear panel **65** defining the rear surface thereof, and the insulation panel **69** between the front panel **61** and the rear panel **65**. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(604) Also, the display **62** may be provided on the rear surface of the front panel **61**. Here, the light guide plate **64** may be provided at the rear of the display **62**. Also, the insulation panel **69** may be omitted if necessary or provided in plurality.

(605) A pair of second spacers **66** may be provided between the rear panel **65**, the insulation panel **69**, and the light guide plate **64** to support the rear panel **65**, the insulation panel **69**, and the light guide plate **64**, and a first spacer **63** may be disposed between the light guide plate **64** and the front panel **61** to support the light guide plate **64** and the front panel **61**. Also, an outer spacer **67** may be disposed outside the first spacer **63** and the second spacers **66**. The outer spacer **67** may be configured to support the front panel **61** and the rear panel **65**. At least one of a space between the first spacer **63** and the second spacer **66** or a space between the outer spacers **67** may be in a vacuum state, or an adiabatic gas may be injected into the space to form an insulation space.

(606) The heater mounting part **673** may be disposed outside the outer spacer **67**. The heater mounting part **673** may protrude outward between the front panel **61** and the rear panel **65**. Also, the heater mounting part **673** may have a height corresponding to the rear panel, and the sealant **608** may be filled between the heater mounting part **673**, the front panel **61**, and the rear panel **65**.

(607) Also, a heater groove **673a** into which the heater **532a** is mounted may be defined in an outer surface of the heater mounting part **673**. The heater groove **673a** may have a size that is enough to accommodate the heater **532a** and be defined in a rear side somewhat than an approximate center or a center of the outer spacer **67**. Thus, in the state in which the transparent panel assembly **60** is disposed on the door **50**, the heater **532a** may be more easily mounted on the outer spacer **67**. Thus, when the heater **532a** is mounted, the heater **532** may not interfere with the constituents within the door **50**. Alternatively, if necessary, in the state in which the heater **532a** is mounted on the periphery of the transparent panel assembly **60**, the transparent panel assembly **60** may be mounted on the door **50**.

(608) When the heater **532a** generates heat, the heat of the heater **532a** may be transferred to the outer spacer **67** made of a metal material. The outer spacer **67** may come into contact with the front panel **61** on a wider area than that of the heater **532a**. Thus, the heat transferred to the outer spacer **67** may heat a wider area when compared with a structure in which the front panel **61** is heated by only the heater **532a**.

(609) Thus, the non-insulation region R between the insulation space of the transparent panel assembly **60** and the insulators **531** of the door **50** may be effectively heated to effectively prevent the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(610) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51** and a panel support part **76** supporting the periphery of the front panel **61**.

(611) In addition to the foregoing embodiment, a refrigerator according to various embodiments

may be exemplified.

(612) In the fifteenth embodiment of the present disclosure, a third spacer and a fourth spacer are provided on the transparent panel assembly, and the heater mounting part and the spacer protrusion are respectively disposed on the third spacer and the fourth spacer.

(613) The fifteenth embodiment is the same as the abovementioned embodiments except for constituents of the transparent panel assembly, and thus, the same constituent as those according to the foregoing embodiments may be denoted by the same reference numeral, and its detailed description will be omitted.

(614) FIG. **61** is a cross-sectional view of a door according to a fifteenth embodiment of the present disclosure.

(615) Referring to the drawing, the door **50** may be provided by coupling the outer plate and the door liner. The transparent panel assembly **60** may be provided in the openings of the outer plate **51** and the door liner **56** so that the interior of the refrigerator **1** is selectively seen. Also, the insulator **531** may be filled into the periphery of the door **50** outside the transparent panel assembly **60**.

(616) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof and the rear panel **65** defining the rear surface thereof. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(617) Also, the display **62** may be provided on the rear surface of the front panel **61**. Here, the light guide plate **64** may be provided at the rear of the display **62**. Also, when the display **62** is omitted, the light guide plate **64** may be omitted, and the insulation panel **69** may be provided. The light guide plate **64** and the insulation panel **69** disposed between the front panel **61** and the rear panel **65** may be intermediate panels, and the intermediate panels may be provided in plurality.

Hereinafter, a structure in which the light guide plate **64** is disposed between the front panel **61** and the rear panel **65** will be described.

(618) A third spacer **663** and a fourth spacer **661** may be provided between the front panel **61**, the light guide plate **64**, and the rear panel **65**. The third spacer **663** and the fourth spacer **661** may be disposed along peripheries of front and rear surfaces of the light guide plate **64** to respectively come into contact with the front panel **61** and the rear panel **65**. Here, the front panel **61** and the rear panel **65** may protrude outward from the third spacer **663** and the fourth spacer **661**, respectively. The front panel **61**, the light guide plate **64**, and the rear panel **65** may be maintained at a set distance by the third spacer **663** and the fourth spacer **661**.

(619) In detail, the third spacer **663** may be disposed between the front panel **61** and the light guide plate **64**. Thus, the light guide plate **64** may be disposed between the front panel **61** and the rear panel **65** to maintain a set distance therebetween and be disposed at suitable distance for visualization of the display **62**.

(620) The heater mounting part **664** may be disposed outside the third spacer **663**. The heater mounting part **664** may protrude outward between the front panel **61** and the light guide plate **64**. The heater mounting part **664** may protrude at a height corresponding to the rear panel **65** to provide a space in which the sealant **608** is applied.

(621) A heater groove **664a** into which the heater **532a** is mounted may be defined in an outer surface of the heater mounting part **664**. The heater groove **664a** may have a size that is enough to accommodate the heater **532a**. Thus, when the heater **532a** generates heat, the heat of the heater **532a** may be transferred to the outer spacer **67** made of a metal material. The outer spacer **67** may come into contact with the front panel **61** on a wider area than that of the heater **532a**. Thus, the heat transferred to the outer spacer **67** may heat a wider area when compared with a structure in which the front panel **61** is heated by only the heater **532a**.

(622) Thus, the non-insulation region R between the insulation space of the transparent panel assembly **60** and the insulators **531** of the door **50** may be effectively heated to effectively prevent

the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(623) The fourth spacer **661** may be disposed between the light guide plate **64** and the rear panel **64** to support the light guide plate **64** and the rear panel **64**. A distance between the light guide plate **64** and the rear panel **65** may be maintained by the fourth spacer **661**.

(624) Also, a spacer protrusion **662** may be disposed outside the fourth spacer **661**. A coupling hole **662a** is defined in an outer surface of the spacer protrusion **662**, and a coupling member **78** passing through the support frame **70** is coupled to the coupling hole **662a**. Thus, the transparent panel assembly **60** may be mounted on the door **50** in the state of being fixed by the support frame **70**.

(625) Also, the spacer protrusion **662** may protrude at a height corresponding to the rear panel **65** and the heater mounting part **664**. Also, the sealant **608** may be filled between the front panel **61** and the rear panel **65** and between the heater mounting part **664** and the spacer protrusion **662**.

(626) Thus, the sealant **608** may be applied at the uniform height on the entire peripheral surface of the transparent panel assembly **60**, and the transparent panel assembly **60** may be more sealed by the sealant **608**. Also, the inside of the sealed transparent panel assembly **60** may be in a vacuum state, and an adiabatic gas may be injected into the inside to form an insulation space.

(627) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51**, a panel support part **76** supporting the periphery of the front panel **61**, and a frame coupling part **77** coupled to the outer spacer **67**.

(628) The frame coupling part **77** may extend upward from an end of the plate support part **74** and also may be disposed or extend further rearward than the spacer protrusion **662**. Thus, the coupling member **78** may be coupled to the frame coupling part **77** so that the transparent panel assembly **60** may be maintained in the state of being fixed and mounted on the inside of the door **50**.

(629) In addition to the foregoing embodiment, a refrigerator according to various embodiments may be exemplified.

(630) In the sixteenth embodiment of the present disclosure, in the transparent panel assembly, a light guide plate and a fifth spacer may be disposed between the front panel and the rear panel, and a distance between the front panel, the rear panel, and the light guide plate may be maintained by the fifth spacer.

(631) The sixteenth embodiment is the same as the abovementioned embodiments except for constituents of the transparent panel assembly, and thus, the same constituent as those according to the foregoing embodiments may be denoted by the same reference numeral, and its detailed description will be omitted.

(632) FIG. **62** is a cross-sectional view of a door according to a sixteenth embodiment of the present disclosure.

(633) Referring to the drawing, the door **50** may be provided by coupling the outer plate **51** and the door liner **56**. The transparent panel assembly **60** may be provided in the openings of the outer plate **51** and the door liner **56** so that the interior of the refrigerator **1** is selectively seen. Also, the insulator **531** may be filled into the periphery of the door **50** outside the transparent panel assembly **60**.

(634) The transparent panel assembly **60** may include the front panel **61** defining the front surface thereof and the rear panel **65** defining the rear surface thereof. A metal deposition layer or a film layer through which light selectively passes may be disposed on the rear surface of the front panel **61**, and thus the interior of the refrigerator **1** may be selectively visualized according to whether the door lights **57** or a lamp in the refrigerator **1** is turned on or off.

(635) Also, the display **62** may be provided on the rear surface of the front panel **61**. Here, the light guide plate **64** may be provided at the rear of the display **62**. Also, when the display **62** is omitted, the light guide plate **64** may be omitted, and the insulation panel **69** may be provided. The light guide plate **64** and the insulation panel **69** disposed between the front panel **61** and the rear panel **65** may be intermediate panels, and the intermediate panels may be provided in plurality.

Hereinafter, a structure in which the light guide plate **64** is disposed between the front panel **61** and the rear panel **65** will be described.

(636) A fifth spacer **666** may be disposed between the front panel **61** and the rear panel **65**. Both ends of the fifth spacer **666** may come into contact with the front panel **61** and the rear panel **65**. Also, a panel groove **667** may be defined in an inner surface of the fifth spacer **666**. The light guide plate **64** may be inserted into the panel groove **667**. The front panel **61**, the light guide plate **64**, and the rear panel **65** may be maintained at a set distance by the fifth spacer **666**.

(637) In detail, the panel groove **667** may be disposed to spaced a predetermined distance from the front panel **61**. Thus, the light guide plate **64** may be disposed between the front panel **61** and the rear panel **65** to maintain a set distance therebetween and be disposed at suitable distance for visualization of the display **62**.

(638) A spacer protrusion **668** may be disposed outside the fifth spacer **666**. The spacer protrusion **668** may protrude from a position that is close to the front panel **61**, i.e., may protrude between the front panel **61** and a heater mounting part **669**. Also, the spacer protrusion **668** may protrude at a height corresponding to the rear panel **65** and the heater mounting part **669**.

(639) A coupling hole **668a** is defined in an outer surface of the spacer protrusion **668**, and a coupling member **78** passing through the support frame **70** is coupled to the coupling hole **668a**. Thus, the transparent panel assembly **60** may be mounted on the door **50** in the state of being fixed by the support frame **70**.

(640) Also, the heater mounting part **669** may be disposed outside the fifth spacer **666**. The heater mounting part **669** may protrude from a position that is close to the rear panel **65**, i.e., may protrude outward between the rear panel **65** and the heater mounting part **669**.

(641) The heater mounting part **669** may protrude at a height corresponding to the rear panel **65**. Thus, in the state in which the sealant **608** is applied, the heater mounting part **669** together with an end of the spacer protrusion **668** may be exposed outward.

(642) A heater groove **669a** into which the heater **532a** is mounted may be defined in an outer surface of the heater mounting part **669**. The heater groove **669a** may have a size that is enough to accommodate the heater **532a**. Thus, when the heater **532a** generates heat, the heat of the heater **532a** may be transferred to the outer spacer **67** made of a metal material. The outer spacer **67** may come into contact with the front panel **61** on a wider area than that of the heater **532a**. Thus, the heat transferred to the outer spacer **67** may heat a wider area when compared with a structure in which the front panel **61** is heated by only the heater **532a**.

(643) Thus, the non-insulation region R between the insulation space of the transparent panel assembly **60** and the insulators **531** of the door **50** may be effectively heated to effectively prevent the dew condensation from being generated on the front surface of the transparent panel assembly **60**.

(644) Each of the spacer protrusion **668** and the heater mounting part **669** may protrude at a height corresponding to the rear panel **65**. Also, the sealant **608** may be filled between the front panel **61** and the rear panel **65** and between the heater mounting part **668** and the spacer protrusion **669**.

(645) Thus, the sealant **608** may be applied at the uniform height on the entire peripheral surface of the transparent panel assembly **60**, and the transparent panel assembly **60** may be more sealed by the sealant **608**. Also, the inside of the sealed transparent panel assembly **60** may be in a vacuum state, and an adiabatic gas may be injected into the inside to form an insulation space.

(646) The support frame **70** may be provided with a plate support part **74** supporting the outer plate **51**, a panel support part **76** supporting the periphery of the front panel **61**, and a frame coupling part **77** coupled to the outer spacer **67**.

(647) The frame coupling part **77** may extend upward from an end of the plate support part **74** and also may be disposed or extend further rearward than the spacer protrusion **668** and also may extend up to a height at which the heater mounting part **669** is covered.

(648) The coupling member **78** may be coupled to the frame coupling part **77** so that the

transparent panel assembly **60** may be maintained in the state of being fixed and mounted on the inside of the door **50**. Also, in the state in which the coupling member **78** coupled to the frame coupling part **77** is coupled to the spacer mounting part **668**, the heater mounting part **669** may be exposed to outside, and thus, the mounting of the heater **523a** may be easily performed. (649) FIG. **63** is a perspective view illustrating a refrigerator according to a seventeenth embodiment of the present disclosure.

(650) As illustrated, a refrigerator **1** according to the tenth embodiment of the present disclosure may be formed by a cabinet **10** in which a storage space is formed, and a plurality of doors **20**, **30**, and **50** configured to open/close the storage space. A first storage space **12** and a second storage space **13** may be vertically partitioned inside the cabinet **10**. Further, the first storage space **12** and the second storage space **13** may be controlled to be operated at different temperatures, and may be configured as, for example, a refrigerating chamber and a freezing chamber. The first storage space **12** and the second storage space **13** may be opened/closed by the pair of doors **20**, **30**, and **50**.

(651) The pair of doors **20** and **50** configured to open/close the first storage space **12** may be rotatably mounted on the cabinet **10**. Further, the pair of doors **20** and **50** may include the door **20** configured to shield the left side of the first storage space **12** and the door **50** configured to shield the right side of the first storage space **12**.

(652) The door **50** may include a see-through part allowing an inside to be selectively seen, and the see-through part may be configured by the transparent panel assembly **60**. Meanwhile, the door **50** may be configured to be identical to any one of the doors according to the above-described embodiments, and the detailed descriptions thereof will be omitted.

(653) Meanwhile, lighting members may be further provided in the door **50** and/or the first storage space **12**, and when the lighting members are turned on, the transparent panel assembly **60** becomes transparent so that a space inside the refrigerator **1** can be seen. Further, when the lighting members are turned off, the transparent panel assembly **60** becomes opaque so that the space inside the refrigerator cannot be seen.

(654) FIG. **64** is a perspective view illustrating a refrigerator according to an eighteenth embodiment of the present disclosure.

(655) As illustrated, a refrigerator **1** according to the eleventh embodiment of the present disclosure may be formed by a cabinet **10** in which a storage space is formed, and a pair of doors **30** and **50** configured to open/close the storage space. A first storage space **12** and a second storage space **13** may be transversely partitioned inside the cabinet **10**. Further, the first storage space **12** and the second storage space **13** may be controlled to be operated at different temperatures, and may be configured as, for example, a refrigerating chamber and a freezing chamber. The first storage space **12** and the second storage space **13** may be opened/closed by the pair of doors **30** and **50**, respectively.

(656) The pair of doors **30** and **50** may be rotatably mounted on the cabinet **10**. Further, the pair of doors **30** and **50** may include the door **30** configured to shield the left second storage space **13** and the door **50** configured to shield the right first storage space **12**.

(657) The door **50** may include a see-through part allowing an inside thereof to be selectively seen, and the see-through part may be configured by the transparent panel assembly **60**. Meanwhile, the door **50** may be configured to be identical to any one of the doors according to the above-described embodiments, and the detailed descriptions thereof will be omitted.

(658) Meanwhile, lighting members may be further provided in the door **50** and/or the first storage space **12**, and when the lighting members are turned on, the transparent panel assembly **60** becomes transparent so that a space inside the refrigerator **1** can be seen. Further, when the lighting members are turned off, the transparent panel assembly **60** becomes opaque so that the space inside the refrigerator cannot be seen.

(659) FIG. **65** is a perspective view illustrating a refrigerator according to a nineteenth embodiment of the present disclosure.

(660) As illustrated, a refrigerator **1** according to the twelfth embodiment of the present disclosure may be formed by a cabinet **10** in which a storage space is formed, and a pair of doors **30** and **50** configured to open/close the storage space. A first storage space **12** and a second storage space **13** may be vertically partitioned inside the cabinet **10**. Further, the first storage space **12** and the second storage space **13** may be controlled to be operated at different temperatures, and may be configured as, for example, a refrigerating chamber and a freezing chamber. The first storage space **12** and the second storage space **13** may be opened/closed by the pair of doors **30** and **50**, respectively.

(661) The pair of doors **30** and **50** may be rotatably mounted on the cabinet **10**. Further, the pair of doors **30** and **50** may include the door **50** configured to shield the first storage space **12** and the door **50** configured to shield the second storage space **13**.

(662) The door **50** may include a see-through part allowing an inside thereof to be selectively seen, and the see-through part may be configured by the transparent panel assembly **60**. Meanwhile, the door **50** may be configured to be identical to any one of the doors according to the above-described embodiments, and the detailed descriptions thereof will be omitted.

(663) Meanwhile, lighting members may be further provided in the door **50** and/or the first storage space **12**, and when the lighting members are turned on, the transparent panel assembly **60** becomes transparent so that a space inside the refrigerator **1** can be seen. Further, when the lighting members are turned off, the transparent panel assembly **60** becomes opaque so that the space inside the refrigerator cannot be seen.

(664) The present disclosure may be applied to all types of refrigerators having a door configured to shield at least a portion of a storage space, regardless of types of refrigerators.

(665) The following effects may be expected in the refrigerator according to the proposed embodiments.

(666) In the refrigerator according to the embodiment of the present disclosure, the see-through part may be selectively switched to be transparent or opaque to visualize the inside of the refrigerator, and the user may check the inside of the refrigerator without opening the door to improve the convenience in use and reduce the power consumption.

(667) The refrigerator according to the embodiment of the present disclosure may have the structure in which the heater is mounted on the outer spacer provided in the transparent panel assembly. Thus, when the heater generates heat, the circumference of the panel may be heated through the outer spacer made of the metal material to prevent the dew condensation from being generated.

(668) Particularly, the outer spacer may have the predetermined thickness to come into contact with the front panel. Thus, when compared to the structure in which the heater comes into contact with the front panel, the wider area may be heated to effectively prevent the dew condensation from being generated.

(669) Also, the outer spacer may be disposed on the insulation space of the transparent panel assembly and the non-insulation region between the insulators around the door. Thus, the outer spacer may generate heat to heat the non-insulation region, thereby preventing the dew condensation from being generated on the front surface of the transparent panel assembly.

(670) Also, the outer spacer may be disposed at the position that is the closest to the visible area of the transparent panel assembly to heat the area adjacent to the visible area without exposing the heater to the outside, thereby effectively preventing the dew condensation from being generated on the visible area.

(671) Also, the heater may be disposed on the outer spacer and be inserted into the heater mounting part that protrudes to the outside. Thus, the heater may be disposed on the outer circumference of the transparent panel assembly so that the heater is disposed through the more simple operation. Also, the heater may be mounted in the state in which the transparent panel assembly is mounted on the door. Therefore, the door may be more easily assembled to improve the assembly

workability and the productivity.

(672) In addition, it may be unnecessary to additionally provide a separate constituent for mounting the heater, and the heater may be mounted on the outer spacer that is previously disposed to more simplify the inner structure of the door, thereby reducing the production cost.

(673) In addition, the heater mounting part on which the heater is mounted may provide the space into which the sealant applied on the circumference of the outer spacer is filled. Therefore, the sealability of the transparent panel assembly may be secured, and the sealant may be uniformly applied.

(674) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A refrigerator comprising: a cabinet defining a storage space; and a door configured to open and close the storage space, the door comprising: an outer plate defining a plate opening and made of a metal material, an upper cap decoration coupled to an upper portion of the outer plate and defining an upper surface of the door, a lower cap decoration coupled to a lower portion of the outer plate and defining a lower surface of the door, a door liner defining a liner opening and coupled to a side portion of the outer plate, a transparent panel assembly that covers the plate opening and the liner opening and is coupled to the outer plate and the door liner, an insulator disposed within a first space defined by the outer plate, the upper cap decoration, the lower cap decoration, the door liner, and the transparent panel assembly, and a gasket coupled to the door liner, wherein the transparent panel assembly comprises: a front panel made of a transparent material, defining a front surface of the door, and including a front protrusion and a bezel, the bezel covering a rear surface of the front protrusion, a rear panel made of a transparent material, spaced apart from the front panel, and defining a rear surface of the door, an outer spacer disposed between the front panel and the rear panel and defining a second space configured to receive gas or maintain a vacuum state, and a sealant disposed between the front panel and the rear panel and configured to cover the outer spacer, and wherein the outer plate comprises: a first part defining a side surface of the door, and a second part extending from the first part and defining the plate opening and a mounting part that is configured to receive the front protrusion of the front panel.

2. The refrigerator according to claim 1, wherein a size of the front panel is greater than a size of the rear panel by an amount of the front protrusion, and wherein the mounting part includes a step recessed with respect to the front surface of the door in a rear direction of the front panel.

3. The refrigerator according to claim 1, wherein the front surface of the door connects the side surface to the mounting part, the front panel being coplanar with the front surface of the door.

4. The refrigerator according to claim 1, further comprising: a blocking part disposed between the insulator and the transparent panel assembly and configured to block a foam liquid from the insulator from permeating along a peripheral surface of the transparent panel assembly.

5. The refrigerator according to claim 4, wherein the blocking part is disposed between the sealant and the insulator and covers the sealant.

6. The refrigerator of claim 4, wherein the sealant defines the peripheral surface of the transparent panel assembly and contacts the blocking part.

7. The refrigerator according to claim 1, wherein the door liner comprises a liner protrusion that

protrudes rearward from the rear panel and that defines a space configured to receive a door light.

8. The refrigerator according to claim 1, further comprising a heater mounted on the front panel, wherein the bezel is provided along a periphery of the front panel and covers the heater.

9. The refrigerator according to claim 1, wherein the bezel is provided through printing using an opaque color to conceal the outer spacer.

10. The refrigerator of claim 1, wherein the transparent panel assembly further comprises an insulation panel disposed between the front panel and the rear panel, and wherein the outer spacer comprises a first spacer that supports the insulation panel and the front panel, and a second spacer that supports the insulation panel and the rear panel.

11. The refrigerator of claim 1, wherein the outer spacer includes a moisture absorbent, and an outer surface of the outer spacer is sealed by the sealant.

12. The refrigerator of claim 1, wherein the door comprises: a main door rotatably coupled to the cabinet and defining an additional storage space, a sub-door rotatably coupled to the main door and having a see-through part, and a lighting module configured to activate to enable visibility inside the additional storage space through the see-through part.

13. The refrigerator of claim 12, wherein the main door comprises an accommodation case provided on the main door and a door basket mounted in the accommodation case, and wherein the additional storage space of the main door is defined by the accommodation case and the door basket.

14. The refrigerator of claim 12, wherein the main door comprises a first handle groove provided at a lower portion of the main door, and wherein the sub-door comprises a second handle groove provided at a lower portion of the sub-door.

15. The refrigerator of claim 14, further comprising an opening device configured to disengage the sub-door from the main door, wherein the opening device comprises: a manipulation lever provided at the lower cap decoration, a rod extending vertically in the first space from the manipulation lever, a lock configured to, based on the lock being moved by the rod when the lock protrudes from a rear surface of the sub-door, release the sub-door from the main door, and a latch hole defined around an opening of the main door, wherein the latch hole is configured to, based on the sub-door closing the opening of the main door, receive the lock.

16. The refrigerator of claim 15, wherein the first handle groove, the second handle groove, and the manipulation lever are disposed at a lower side of the see-through part.

17. The refrigerator of claim 12, wherein the transparent panel assembly further comprises: a display disposed within the second space and configured to output screen information, a light guide plate disposed between the display and the rear panel within the second space, and a display light provided at a side of the light guide plate within the second space and configured to emit light to the light guide plate.

18. The refrigerator of claim 17, wherein the transparent panel assembly further comprises a cable connected to the display, arranged along a periphery of the outer spacer, and passing through the sealant.

19. The refrigerator of claim 18, further comprising a control unit configured to control the lighting module, the display, and the display light, wherein the see-through part is configured to operate in a plurality of mode states, the plurality of mode states comprising: a transparent mode state in which the lighting module is activated to enable the storage space to be seen through the transparent panel assembly, while the display light is turned off and the display does not output the screen information, an opaque mode state in which the lighting module is deactivated to render the storage space concealed behind the transparent panel assembly, while the display light is turned off and the display does not output the screen information, and a screen output mode state in which the lighting module is deactivated and the display light is activated, the display being configured to output the screen information through the transparent panel assembly.

20. The refrigerator of claim 1, wherein the storage space comprises a refrigerating space disposed

in an upper portion of the cabinet and a freezing space disposed in a lower portion of the cabinet, wherein a first door and a second door are configured to open and close the refrigerating space, and wherein the transparent panel assembly is disposed at one of the first door or the second door to enable an inside of the refrigerating space to be seen through the transparent panel assembly.
